

The equations $\sin x = a$, $\cos x = a$, $\tan x = a$, $\cot x = a$

One arc function gives one solution; the symmetry of the circle gives all the rest.

LESSON 79–81

3 × 40 MIN

ALGEBRA 10, §26

P1 5.1, 5.4

LESSON CLOCK

12

30'

30'

35'

22'

6

When there is a solution at all	12 min
The four formulas	30 min
Reading them off the circle	30 min
Special values	35 min
Solutions in an interval	22 min
Homework	6 min

LEARNING OBJECTIVES

- Decide when $\sin x = a$ and $\cos x = a$ have solutions.
- Write the general solution of each of the four basic equations.
- Read every solution off the unit circle rather than memorising formulas.
- Select the solutions that lie in a given interval.

TEXTBOOK

National	pp. 258–270
Cambridge	Pure Mathematics 1, pp. 96–104
Workbook	Exercise 5D

01

Explanation

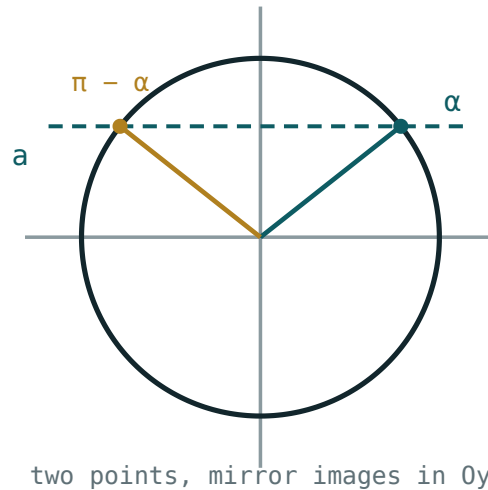
When there is a solution at all

Before any formula, one question: can the equation hold?

EQUATION	RANGE OF THE LEFT SIDE	SOLVABLE WHEN
$\sin x = a$	$[-1, 1]$	$ a \leq 1$
$\cos x = a$	$[-1, 1]$	$ a \leq 1$
$\tan x = a$	$\mathbb{R}$	always
$\cot x = a$	$\mathbb{R}$	always

WRITE “NO SOLUTION”, AND SAY WHY

$\sin x = 1.4$ has no solution because the sine never leaves $[-1, 1]$. A blank page earns nothing; one sentence earns the mark.



Two points on the circle at the same height — the two families of solutions of $\sin x = a$.

The four formulas

$$\sin x = a \Rightarrow x = (-1)^n \arcsin a + \pi n, n \in \mathbb{Z}$$

$$\cos x = a \Rightarrow x = \pm \arccos a + 2\pi n, n \in \mathbb{Z}$$

$$\tan x = a \Rightarrow x = \arctan a + \pi n, n \in \mathbb{Z}$$

$$\cot x = a \Rightarrow x = \operatorname{arccot} a + \pi n, n \in \mathbb{Z}$$

Three things are worth noticing at once.

- The sine and cosine formulas step by 2π but pick up a second family; the compact $(-1)^n$ form packs both into one line, alternating between α and $\pi - \alpha$.
- The tangent and cotangent step by π , because their period is π , not 2π . One family only.
- Every formula begins with an arc function, so its **principal value** is the seed and the period does the rest.

THE TWO SPECIAL-CASE SHORTCUTS

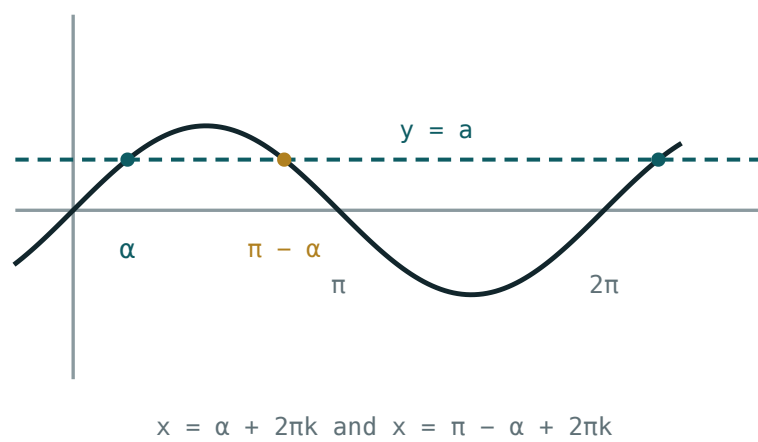
For $a = 0, \pm 1$ the general formula still works, but the direct answer is shorter: $\sin x = 0 \Rightarrow x = \pi n$; $\sin x = 1 \Rightarrow x = \frac{\pi}{2} + 2\pi n$; $\cos x = 0 \Rightarrow x = \frac{\pi}{2} + \pi n$; $\cos x = 1 \Rightarrow x = 2\pi n$.

Reading them off the circle

Do not memorise the four lines. Draw the circle, mark the horizontal or vertical line at height a , and read the two points.

EQUATION	DRAW	THE TWO POINTS
$\sin x = a$	the horizontal line $y = a$	α and $\pi - \alpha$
$\cos x = a$	the vertical line $x = a$	α and $-\alpha$
$\tan x = a$	the line through O of gradient a	α and $\alpha + \pi$

The tangent's two points are **opposite** ends of one diameter, which is exactly why the step is π . The sine's are mirror images in the vertical axis, the cosine's in the horizontal axis. Three pictures, four formulas, nothing to memorise.



The same equation on the graph: $y = a$ cuts the sine curve twice in every period.

Special values

Almost every examination equation uses one of these.

A	$\text{ARCSIN } A$	$\text{ARCCOS } A$	$\text{ARCTAN } A$
0	0	$\frac{\pi}{2}$	0
$\frac{1}{2}$	$\frac{\pi}{6}$	$\frac{\pi}{3}$	—
$\frac{\sqrt{2}}{2}$	$\frac{\pi}{4}$	$\frac{\pi}{4}$	—
$\frac{\sqrt{3}}{2}$	$\frac{\pi}{3}$	$\frac{\pi}{6}$	—
1	$\frac{\pi}{2}$	0	$\frac{\pi}{4}$
$\sqrt{3}$	—	—	$\frac{\pi}{3}$

A NEGATIVE A IS NOT SYMMETRIC FOR THE COSINE

$\arcsin(-a) = -\arcsin a$ and $\arctan(-a) = -\arctan a$, but $\arccos(-a) = \pi - \arccos a$. So $\arccos(-\frac{1}{2}) = \frac{2\pi}{3}$, not $-\frac{\pi}{3}$.

Solutions in an interval

An examination rarely wants the general solution. It wants the solutions in, say, $[0, 2\pi]$ or $[-180^\circ, 180^\circ]$. Method:

- 1. write the general solution;
- 2. substitute $n = \dots, -1, 0, 1, 2, \dots$ one at a time;
- 3. keep each value that falls in the interval, and stop when they leave it.

Example. $\cos x = \frac{1}{2}$ on $[0, 2\pi]$. General: $x = \pm \frac{\pi}{3} + 2\pi n$. Then:

N	$+ \text{BRANCH}$	$- \text{BRANCH}$
0	$\frac{\pi}{3} \checkmark$	$-\frac{\pi}{3} \times$
1	$\frac{7\pi}{3} \times$	$\frac{5\pi}{3} \checkmark$

So $x = \frac{\pi}{3}$ and $x = \frac{5\pi}{3}$. Two solutions, as the picture predicted: a horizontal chord meets the circle twice.

A COUNT YOU CAN CHECK AGAINST

On an interval of length 2π , $\sin x = a$ and $\cos x = a$ have **two** solutions when $|a| < 1$, one when $|a| = 1$; $\tan x = a$ has **two**, one in each half-turn. If your count differs, you have lost a family.

02 Worked examples

EXAMPLE 1 Solve $\sin x = \frac{1}{2}$ in general, then on $[0, 2\pi]$.

① $\arcsin \frac{1}{2} = \frac{\pi}{6}$

The seed.

② $x = (-1)^n \frac{\pi}{6} + \pi n$

General.

③ $n = 0: \frac{\pi}{6}; n = 1: \pi - \frac{\pi}{6} = \frac{5\pi}{6}$

④ $n = 2: 2\pi + \frac{\pi}{6} \text{ — outside.}$

ANSWER

$$x = \frac{\pi}{6} \text{ and } x = \frac{5\pi}{6}$$

EXAMPLE 2 Solve $\cos x = -\frac{\sqrt{3}}{2}$.

$$\textcircled{1} \quad \arccos\left(-\frac{\sqrt{3}}{2}\right) = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

Not $-\frac{\pi}{6}$.

$$\textcircled{2} \quad x = \pm \frac{5\pi}{6} + 2\pi n$$

$$\textcircled{3} \quad \text{On } [0, 2\pi]: \frac{5\pi}{6} \text{ and } \frac{7\pi}{6}.$$

ANSWER

$$x = \pm \frac{5\pi}{6} + 2\pi n$$

EXAMPLE 3 Solve $\tan x = -1$ on $[0^\circ, 360^\circ]$.

$$\textcircled{1} \quad \arctan(-1) = -45^\circ$$

$$\textcircled{2} \quad x = -45^\circ + 180^\circ n$$

Step 180° , one family.

$$\textcircled{3} \quad n = 1: 135^\circ; n = 2: 315^\circ$$

ANSWER

135° and 315°

EXAMPLE 4 Solve $\sin 2x = \frac{\sqrt{2}}{2}$ on $[0, \pi]$.

① Let $t = 2x$, so $t \in [0, 2\pi]$.

Substitute, then widen the interval.

② $t = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{9\pi}{4}$ ✗

③ $x = \frac{\pi}{8}, \frac{3\pi}{8}$

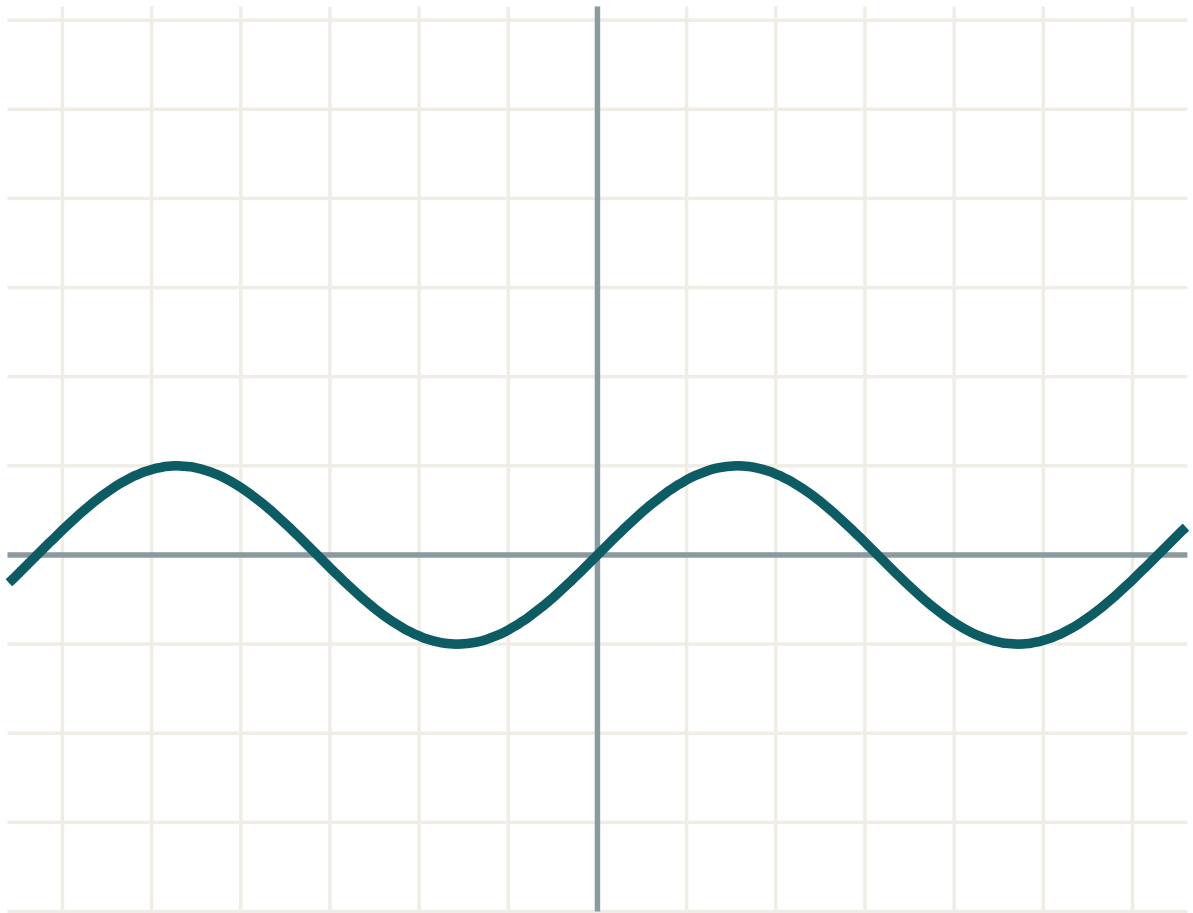
Halve each.

ANSWER

$$x = \frac{\pi}{8} \text{ and } x = \frac{3\pi}{8}$$

03 Interactive model

Draw one circle on the board and solve all four equations on it, rubbing out only the chord.

y = a against the sine curve

a 1

b 0

c 0

equation $y = f(x)$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Trigonometric equation	Trigonometrik tenglama	Тригонометрическое уравнение
General solution	Umumiy yechim	Общее решение
Principal value	Bosh qiymat	Главное значение
Arcsine	Arksinus	Арксинус
Period	Davr	Период
Unit circle	Birlik aylana	Единичная окружность
Solution set	Yechimlar to'plami	Множество решений
Interval	Oraliq	Промежуток
Reference angle	Tayanch burchak	Опорный угол
Integer parameter	Butun parametr	Целый параметр

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$\sin x = 1.2$ has:

two solutions

one solution

no solution

infinitely many

The period in the tangent formula is:

2π

π

$\frac{\pi}{2}$

4π

$\arccos(-\frac{1}{2})$ equals:

$-\frac{\pi}{3}$

$\frac{2\pi}{3}$

$\frac{\pi}{3}$

$\frac{5\pi}{6}$

$\cos x = a$ with $|a| < 1$ has, on $[0, 2\pi]$:

 one solution

 two solutions

 three

 four

The seed of every general solution is:

 zero

 the arc function

 the period

 π

$\sin x = -1$ gives:

$x = -\frac{\pi}{2} + 2\pi n$

$x = \pm \frac{\pi}{2}$

$x = \pi n$

 no solution

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $\sin x = 0$

ANSWER $x = \pi n$

2. Solve $\cos x = 1$

ANSWER $x = 2\pi n$

3. Solve $\cos x = 0$

ANSWER $x = \frac{\pi}{2} + \pi n$

4. Solve $\tan x = 0$

ANSWER $x = \pi n$

5. Solve $\sin x = 1$

ANSWER $x = \frac{\pi}{2} + 2\pi n$

6. Does $\cos x = -1.5$ have a solution?

ANSWER No — $|a| > 1$

7. Solve $\tan x = 1$

ANSWER $x = \frac{\pi}{4} + \pi n$

Medium · the standard question

7 PROBLEMS

1. Solve $\sin x = \frac{\sqrt{2}}{2}$

ANSWER $x = (-1)^n \frac{\pi}{4} + \pi n$

2. Solve $\cos x = \frac{1}{2}$

ANSWER $x = \pm \frac{\pi}{3} + 2\pi n$

3. Solve $\tan x = \sqrt{3}$

ANSWER $x = \frac{\pi}{3} + \pi n$

4. Solve $\cos x = -1$

ANSWER $x = \pi + 2\pi n$

5. Solve $\sin x = \frac{1}{2}$ on $[0, 2\pi]$

ANSWER $\frac{\pi}{6}, \frac{5\pi}{6}$

6. Solve $\cos x = \frac{\sqrt{3}}{2}$ on $[0^\circ, 360^\circ]$

ANSWER $30^\circ, 330^\circ$

7. Solve $\tan x = -\frac{\sqrt{3}}{3}$ on $[0^\circ, 360^\circ]$

ANSWER $150^\circ, 330^\circ$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\sin 2x = \frac{\sqrt{3}}{2}$ on $[0, \pi]$

ANSWER $\frac{\pi}{6}, \frac{\pi}{3}$

2. Solve $\cos(x - \frac{\pi}{4}) = \frac{1}{2}$

ANSWER $x = \frac{\pi}{4} \pm \frac{\pi}{3} + 2\pi n$

3. Solve $\tan 3x = 1$ on $[0, \pi]$

ANSWER $\frac{\pi}{12}, \frac{5\pi}{12}, \frac{3\pi}{4}$

4. Solve $\sin(2x + \frac{\pi}{3}) = 0$

ANSWER $x = -\frac{\pi}{6} + \frac{\pi n}{2}$

5. How many solutions has $\cos 3x = \frac{1}{2}$ on $[0, 2\pi]$?

ANSWER 6

6. Solve $2 \sin x - 1 = 0$ on $[-\pi, \pi]$

ANSWER $\frac{\pi}{6}, \frac{5\pi}{6}$

7. For which a has $\sin x = a$ exactly one solution on $[0, 2\pi)$?

ANSWER $a = \pm 1$

07 Homework

Homework — 5 tasks

Draw the circle for every question before writing a formula.

1. Solve $\sin x = -\frac{\sqrt{2}}{2}$ in general and on $[0, 2\pi]$.
2. Solve $\cos x = -\frac{1}{2}$ in general and on $[0^\circ, 360^\circ]$.
3. Solve $\tan x = \sqrt{3}$ on $[-180^\circ, 180^\circ]$.
4. Solve $\sin 3x = \frac{1}{2}$ on $[0, \pi]$, showing the substitution $t = 3x$.
5. Explain in two sentences why $\tan x = a$ always has a solution but $\sin x = a$ may not.

Methods for solving trigonometric equations

Factorise, substitute, or reduce to one function — three moves cover almost everything.

LESSON 82–85

4 × 40 MIN

ALGEBRA 10, §27

P1 5.6–5.7

LESSON CLOCK

20'

40'

40'

40'

35'

5

The three moves	20 min
Quadratic by substitution	40 min
Factorising	40 min
Homogeneous equations	40 min
Double angle and auxiliary angle	35 min
Homework	5 min

LEARNING OBJECTIVES

- Reduce an equation to a single trigonometric function.
- Solve quadratic equations in $\sin x$ or $\cos x$ by substitution.
- Solve homogeneous equations by dividing by $\cos x$.
- Use the identities for $\sin 2x$, $\cos 2x$ and $a \sin x + b \cos x$.

TEXTBOOK

National	pp. 271–288
Cambridge	Pure Mathematics 1, pp. 105–116
Workbook	Exercise 5E–5F

01

Explanation

The three moves

Every equation in this section is turned into $\sin x = a$ or one of its three relatives. There are only three ways to get there.

MOVE	RECOGNISE IT BY	EXAMPLE
substitute	one function, two powers	$2\sin^2x - 3\sin x + 1 = 0$
factorise	a common factor, or a product = 0	$\sin x \cos x - \sin x = 0$
reduce	two functions of the same angle	$2\cos^2x + \sin x = 2$

THE IDENTITY THAT DOES MOST OF THE WORK

$\sin^2x + \cos^2x = 1$

Read left to right it removes a $\cos^2$; read right to left it creates one. Almost every “reduce” step is this identity used in one direction or the other.

Quadratic by substitution

Example. $2\sin^2 x - 3\sin x + 1 = 0$. Put $t = \sin x$:

$$2t^2 - 3t + 1 = 0 \Rightarrow t = 1 \text{ or } t = \frac{1}{2}$$

Now solve each: $\sin x = 1 \Rightarrow x = \frac{\pi}{2} + 2\pi n$, and $\sin x = \frac{1}{2} \Rightarrow x = (-1)^n \frac{\pi}{6} + \pi n$.

CHECK EVERY ROOT AGAINST $|T| \leq 1$

$2\sin^2 x - \sin x - 6 = 0$ gives $t = 2$ or $t = -\frac{3}{2}$. Both are outside $[-1, 1]$, so the equation has **no** solutions. Do the check before solving.

When the equation contains $\sin^2 x$ and $\cos x$, replace first: $2\cos^2 x + \sin x = 2$ becomes $2(1 - \sin^2 x) + \sin x = 2$, that is $2\sin^2 x - \sin x = 0$ — now a quadratic in $\sin x$ that also factorises.

Factorising

When every term shares a factor, take it out — **never** divide by it.

$$\sin x \cos x = \sin x \Rightarrow \sin x (\cos x - 1) = 0$$

So $\sin x = 0$ or $\cos x = 1$, giving $x = \pi n$ together with $x = 2\pi n$ — and the second family is inside the first, so the answer is just $x = \pi n$.

DIVIDING BY $\sin x$ LOSES SOLUTIONS

Divide $\sin x \cos x = \sin x$ by $\sin x$ and you get $\cos x = 1$ alone — every solution with $\sin x = 0$ has vanished. Factor, set each factor to zero, keep both.

EQUATION	FACTORED	SOLUTIONS
$2\sin x \cos x = \cos x$	$\cos x(2\sin x - 1) = 0$	$\frac{\pi}{2} + \pi n; (-1)^n \frac{\pi}{6} + \pi n$
$\tan x \cdot \sin x = \tan x$	$\tan x(\sin x - 1) = 0$	$\pi n; \frac{\pi}{2} + 2\pi n$
$\sin 2x = \sin x$	$\sin x(2\cos x - 1) = 0$	$\pi n; \pm \frac{\pi}{3} + 2\pi n$

Homogeneous equations

An equation is **homogeneous of degree one** when every term has exactly one factor of $\sin x$ or $\cos x$:

$$a \sin x + b \cos x = 0$$

Here dividing is safe — by $\cos x$, because $\cos x = 0$ would force $\sin x = 0$ too, and the two are never zero together. The result:

$$a \tan x + b = 0 \Rightarrow \tan x = -\frac{b}{a}$$

Degree two works the same way. $3\sin^2 x - 4\sin x \cos x + \cos^2 x = 0$: divide through by $\cos^2 x$ to get $3\tan^2 x - 4\tan x + 1 = 0$, a quadratic in $\tan x$ with roots 1 and $\frac{1}{3}$.

HOMOGENEOUS, OR NOT?

Count the trigonometric factors in each term. $3\sin^2 x - 4\sin x \cos x + \cos^2 x$ — two, two, two: homogeneous. $3\sin^2 x - 4\sin x \cos x + 1$ — two, two, **zero**: not homogeneous, until you write $1 = \sin^2 x + \cos^2 x$ and it becomes so.

Double angle and the auxiliary angle

$$\sin 2x = 2 \sin x \cos x \quad \cos 2x = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x = 2\cos^2 x - 1$$

Three forms of $\cos 2x$, and the right one is the one that leaves a single function. $\cos 2x + \sin x = 0$ becomes $1 - 2\sin^2 x + \sin x = 0$, a quadratic in $\sin x$.

The auxiliary angle. An equation of the form $a \sin x + b \cos x = c$ is not homogeneous, because of the constant. Write the left side as a single sine:

$$a \sin x + b \cos x = R \sin(x + \alpha), R = \sqrt{a^2 + b^2}, \tan \alpha = \frac{b}{a}$$

Example. $\sqrt{3} \sin x + \cos x = 1$. Here $R = 2$ and $\alpha = \frac{\pi}{6}$, so $2 \sin(x + \frac{\pi}{6}) = 1$, that is

$$\sin(x + \frac{\pi}{6}) = \frac{1}{2}.$$

WHAT R TELLS YOU IMMEDIATELY

Since $|\sin| \leq 1$, the expression $a \sin x + b \cos x$ never leaves $[-R, R]$. So $\sqrt{3} \sin x + \cos x = 3$ has no solution — $R = 2 < 3$ — and the maximum of $3 \sin x + 4 \cos x$ is 5. One number answers a whole family of questions.

02 Worked examples

EXAMPLE 1 Solve $2\cos^2 x + 3 \sin x = 0$.

① $2(1 - \sin^2 x) + 3 \sin x = 0$
Reduce to one function.

② $2\sin^2 x - 3 \sin x - 2 = 0$

③ $t = 2$ (rejected) or $t = -\frac{1}{2}$
 $|t| \leq 1$ kills the first.

④ $x = (-1)^{n+1} \frac{\pi}{6} + \pi n$

ANSWER

$$x = (-1)^{n+1} \frac{\pi}{6} + \pi n$$

EXAMPLE 2 Solve $\sin 2x = \cos x$.

① $2 \sin x \cos x - \cos x = 0$
Double angle first.

② $\cos x (2 \sin x - 1) = 0$
Factor — do not divide.

③ $\cos x = 0 \Rightarrow x = \frac{\pi}{2} + \pi n$

④ $\sin x = \frac{1}{2} \Rightarrow x = (-1)^n \frac{\pi}{6} + \pi n$

ANSWER

$$x = \frac{\pi}{2} + \pi n \text{ or } x = (-1)^n \frac{\pi}{6} + \pi n$$

EXAMPLE 3 Solve $\sin^2 x - 3 \sin x \cos x + 2 \cos^2 x = 0$.

① Homogeneous of degree 2; divide by $\cos^2 x$.

② $\tan^2 x - 3 \tan x + 2 = 0$

③ $\tan x = 1$ or $\tan x = 2$

④ $x = \frac{\pi}{4} + \pi n$ or $x = \arctan 2 + \pi n$

ANSWER

$$x = \frac{\pi}{4} + \pi n, x = \arctan 2 + \pi n$$

EXAMPLE 4 Solve $\sqrt{3} \sin x + \cos x = 1$ on $[0, 2\pi]$.

$$① \quad R = \sqrt{3+1} = 2, \tan \alpha = \frac{1}{\sqrt{3}} \Rightarrow \alpha = \frac{\pi}{6}$$

$$② \quad 2 \sin\left(x + \frac{\pi}{6}\right) = 1$$

$$③ \quad x + \frac{\pi}{6} = \frac{\pi}{6} \text{ or } \frac{5\pi}{6} \pmod{2\pi}$$

$$④ \quad x = 0 \text{ or } x = \frac{2\pi}{3}; \text{ also } x = 2\pi.$$

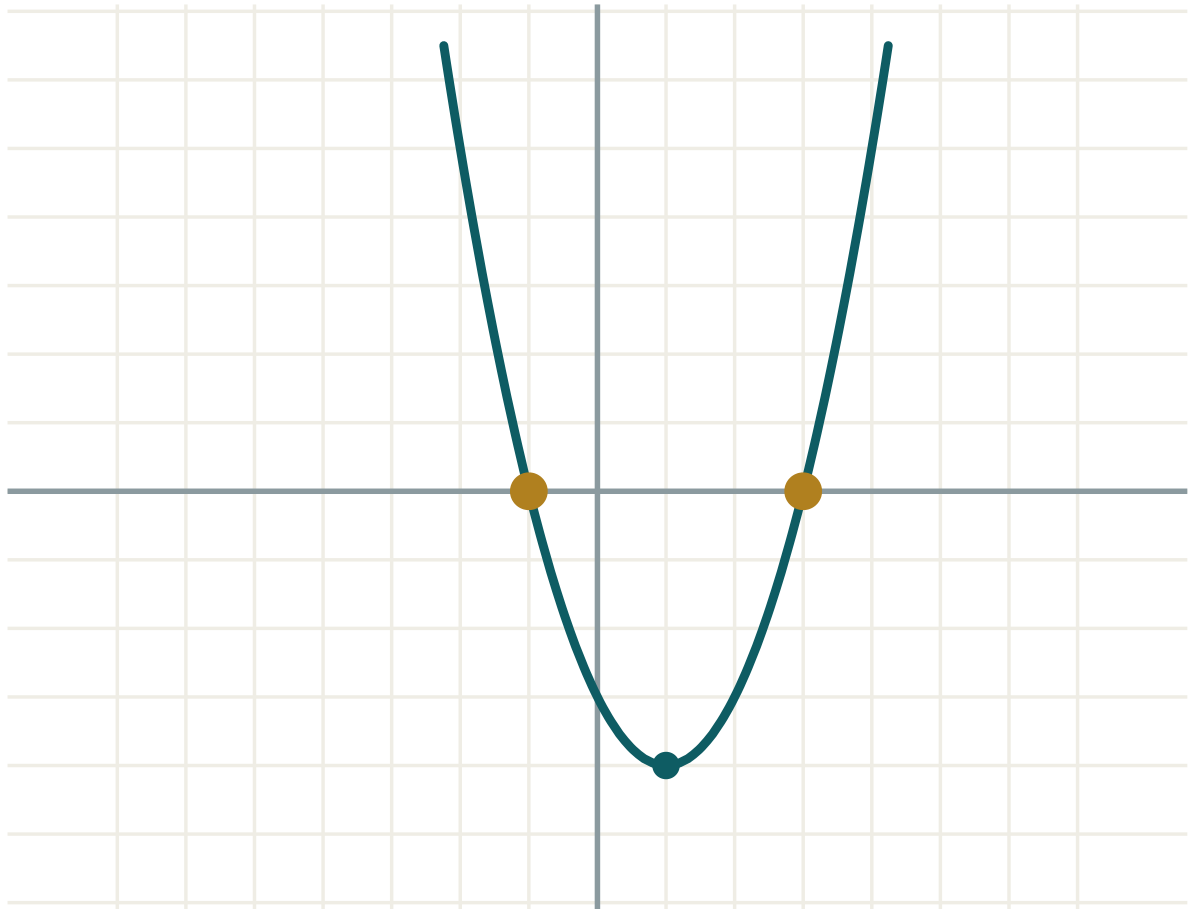
ANSWER

$$x = 0, \frac{2\pi}{3}, 2\pi$$

03 Interactive model

Put the three moves on the board as a checklist and refuse to start any question until the class has named the move.

The quadratic behind the substitution



a 1

b -2

c -3

 $D = b^2 - 4ac$ 16

roots two

 x_1 -1 x_2 3 $x_1 + x_2$ 2 $x_1 \cdot x_2$ -3

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Substitution	O'rniga qo'yish	Подстановка
Homogeneous equation	Bir jinsli tenglama	Однородное уравнение
Factorisation	Ko'paytuvchilarga ajratish	Разложение на множители
Double angle	Ikkilangan burchak	Двойной угол
Identity	Ayniyat	Тождество
Auxiliary angle	Yordamchi burchak	Вспомогательный угол
Extraneous root	Chet ildiz	Посторонний корень
Domain restriction	Aniqlanish sohasi cheklovi	Ограничение области
Quadratic in $\sin x$	$\sin x$ ga nisbatan kvadrat	Квадратное относительно $\sin x$
Reduction	Keltirish	Приведение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$2\sin^2x - \sin x - 6 = 0$ has:

two families

one family

no solution

four families

To solve $\sin x \cos x = \sin x$ you should:

divide by $\sin x$

factorise

square both sides

substitute $t = \cos x$

A homogeneous equation is solved by dividing by:

$\sin x$

the highest power of $\cos x$

2

nothing

The maximum of $3 \sin x + 4 \cos x$ is:

Which form of $\cos 2x$ suits $\cos 2x + \sin x = 0$?

$a \sin x + b \cos x = c$ has no solution when:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $\sin x (\cos x - 1) = 0$

ANSWER $x = \pi n$

2. Solve $\cos x (2 \sin x - 1) = 0$

ANSWER $\frac{\pi}{2} + \pi n; (-1)^n \frac{\pi}{6} + \pi n$

3. Substitute $t = \sin x$ in $2\sin^2 x - 3 \sin x + 1 = 0$

ANSWER $2t^2 - 3t + 1 = 0$

4. Roots of $2t^2 - 3t + 1 = 0$

ANSWER $t = 1, t = \frac{1}{2}$

5. Replace $\cos^2 x$ in $2\cos^2 x + \sin x = 2$

ANSWER $2\sin^2 x - \sin x = 0$

6. R for $3 \sin x + 4 \cos x$

ANSWER 5

7. R for $\sin x + \cos x$

ANSWER $\sqrt{2}$

Medium · the standard question

7 PROBLEMS

1. Solve $2\sin^2 x - 3 \sin x + 1 = 0$

ANSWER $\frac{\pi}{2} + 2\pi n; (-1)^n \frac{\pi}{6} + \pi n$

2. Solve $2\cos^2 x + \sin x = 2$

ANSWER $\pi n; (-1)^n \frac{\pi}{6} + \pi n$

3. Solve $\sin 2x = \sin x$

ANSWER $\pi n; \pm \frac{\pi}{3} + 2\pi n$

4. Solve $\sin x + \cos x = 0$

ANSWER $x = -\frac{\pi}{4} + \pi n$

5. Solve $\sqrt{3} \sin x - \cos x = 0$

ANSWER $x = \frac{\pi}{6} + \pi n$

6. Solve $\cos 2x + \cos x = 0$ on $[0, 2\pi]$

ANSWER $\frac{\pi}{3}, \pi, \frac{5\pi}{3}$

7. Solve $\tan^2 x - 1 = 0$

ANSWER $x = \pm \frac{\pi}{4} + \pi n$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\sin^2 x - 3 \sin x \cos x + 2 \cos^2 x = 0$

ANSWER $\frac{\pi}{4} + \pi n; \arctan 2 + \pi n$

2. Solve $\sqrt{3} \sin x + \cos x = 1$ on $[0, 2\pi]$

ANSWER $0, \frac{2\pi}{3}, 2\pi$

3. Solve $\sin x + \cos x = 1$ on $[0, 2\pi]$

ANSWER $0, \frac{\pi}{2}, 2\pi$

4. Solve $\cos 2x = 3 \cos x - 2$ on $[0, 2\pi]$

ANSWER $0, \frac{\pi}{3}, \frac{5\pi}{3}, 2\pi$

5. Solve $\sin 2x + \cos 2x = 1$ on $[0, 2\pi]$

ANSWER $0, \frac{\pi}{4}, \pi, \frac{5\pi}{4}, 2\pi$

6. Find the maximum and minimum of $5 \sin x - 12 \cos x$

ANSWER 13 and -13

7. For which c has $\sin x + \sqrt{3} \cos x = c$ a solution?

ANSWER $|c| \leq 2$

07 Homework

Homework — 5 tasks

Name the move — substitute, factorise or reduce — before each solution.

1. Solve $2\sin^2 x + \sin x - 1 = 0$ in general and on $[0, 2\pi]$.
2. Solve $\sin 2x = \cos x$ and explain in one sentence why dividing by $\cos x$ would be wrong.
3. Solve $3\sin^2 x - 4 \sin x \cos x + \cos^2 x = 0$.

4. Write $5 \sin x + 12 \cos x$ in the form $R \sin(x + \alpha)$ and state its greatest value.
5. Solve $\cos 2x + 3 \sin x = 2$ on $[0^\circ, 360^\circ]$.

Trigonometric inequalities

Solve the equation first, then read the arc between the two points.

LESSON 86–88

3 × 40 MIN

ALGEBRA 10, §28

EXTENSION BEYOND P1

LESSON CLOCK

15'

30'

30'

25'

30'

5

From points to arcs	15 min
$\sin x > a$	30 min
$\cos x > a$	30 min
tangent inequalities	25 min
Quadratic and compound	30 min
Homework	5 min

LEARNING OBJECTIVES

- Solve $\sin x > a$ and its three relatives on the unit circle.
- Write the answer as a family of intervals, not of points.
- Choose the arc correctly for a negative right-hand side.
- Solve simple compound and quadratic trigonometric inequalities.

TEXTBOOK

National	pp. 289–300
Cambridge	Pure Mathematics 1, pp. 117–120
Workbook	Exercise 5G

01

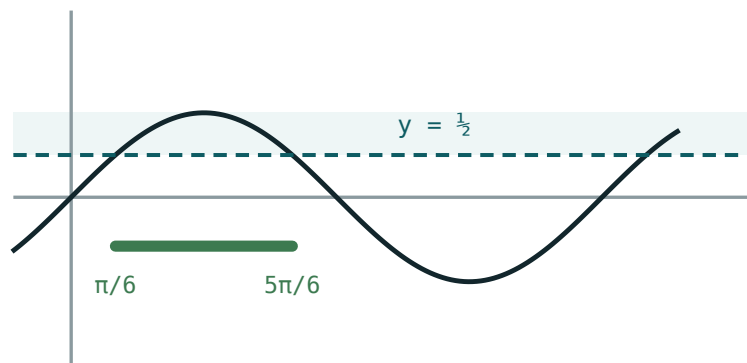
Explanation

From points to arcs

The equation $\sin x = a$ marks two points on the circle. The inequality $\sin x > a$ asks for the **arc** between them — the part of the circle lying above the chord.

THE METHOD, IN THREE LINES

- Solve the equation to find the two boundary points.
- Shade the arc on which the inequality is true.
- Read the arc from its start to its end, going **anticlockwise**, and add the period.



$$\sin x > \frac{1}{2} \Leftrightarrow \pi/6 + 2\pi k < x < 5\pi/6 + 2\pi k$$

$\sin x > \frac{1}{2}$ — the band above the line, and the intervals it cuts.

ALWAYS GO ANTICLOCKWISE

Read the arc in the direction of increasing angle. Written the other way, $\frac{5\pi}{6} < x < \frac{\pi}{6}$ is an empty interval, and the whole answer collapses.

$\sin x > a$ and $\sin x < a$

For $\sin x > a$, the arc runs from $\arcsin a$ anticlockwise to $\pi - \arcsin a$ — the upper arc.

$$\sin x > a \Leftrightarrow \arcsin a + 2\pi n < x < \pi - \arcsin a + 2\pi n$$

$$\sin x < a \Leftrightarrow \pi - \arcsin a + 2\pi n < x < 2\pi + \arcsin a + 2\pi n$$

INEQUALITY	BOUNDARY POINTS	ANSWER ON ONE PERIOD
$\sin x > \frac{1}{2}$	$\frac{\pi}{6}, \frac{5\pi}{6}$	$(\frac{\pi}{6}, \frac{5\pi}{6})$
$\sin x \geq \frac{\sqrt{2}}{2}$	$\frac{\pi}{4}, \frac{3\pi}{4}$	$[\frac{\pi}{4}, \frac{3\pi}{4}]$
$\sin x < \frac{1}{2}$	$\frac{\pi}{6}, \frac{5\pi}{6}$	$(\frac{5\pi}{6}, \frac{13\pi}{6})$
$\sin x > -\frac{1}{2}$	$-\frac{\pi}{6}, \frac{7\pi}{6}$	$(-\frac{\pi}{6}, \frac{7\pi}{6})$

Notice that the last line has the **longer** arc: a negative bound leaves most of the circle above the chord. If your interval for $\sin x > -\frac{1}{2}$ is shorter than half the circle,

you have shaded the wrong arc.

$\cos x > a$ and $\cos x < a$

Here the chord is **vertical**, and the boundary points are $\pm \arccos a$. The arc where the cosine is large lies on the right.

$$\cos x > a \Leftrightarrow -\arccos a + 2\pi n < x < \arccos a + 2\pi n$$

$$\cos x < a \Leftrightarrow \arccos a + 2\pi n < x < 2\pi - \arccos a + 2\pi n$$

INEQUALITY	ANSWER ON ONE PERIOD	LENGTH OF THE ARC
$\cos x > \frac{1}{2}$	$(-\frac{\pi}{3}, \frac{\pi}{3})$	$\frac{2\pi}{3}$
$\cos x < \frac{1}{2}$	$(\frac{\pi}{3}, \frac{5\pi}{3})$	$\frac{4\pi}{3}$
$\cos x \geq -\frac{\sqrt{2}}{2}$	$[-\frac{3\pi}{4}, \frac{3\pi}{4}]$	$\frac{3\pi}{2}$

A CHECK THAT COSTS NOTHING

The two arcs must add to 2π . If $\cos x > \frac{1}{2}$ gives $\frac{2\pi}{3}$ and $\cos x < \frac{1}{2}$ gives $\frac{4\pi}{3}$, the pair is consistent.

Tangent inequalities

The tangent has period π and a vertical asymptote in the middle of each period, so its solution intervals are **half** as long and never cross the asymptote.

$$\tan x > a \Leftrightarrow \arctan a + \pi n < x < \frac{\pi}{2} + \pi n$$

$$\tan x < a \Leftrightarrow -\frac{\pi}{2} + \pi n < x < \arctan a + \pi n$$

The interval is always **open** at $\frac{\pi}{2} + \pi n$, because the tangent is not defined there.

Writing $\leq \frac{\pi}{2}$ is a real error, not a small one.

THE PERIOD IS π , NOT 2π

Adding $2\pi n$ to a tangent inequality halves the solution set. Every tangent answer steps by π .

Quadratic and compound inequalities

Substitute exactly as for equations, solve the quadratic in t , then intersect with $-1 \leq t \leq 1$ and translate back.

Example. $2\sin^2x - \sin x - 1 > 0$. With $t = \sin x$: $(2t + 1)(t - 1) > 0$, so $t < -\frac{1}{2}$ or $t > 1$

1. The second is impossible, so the answer comes from $\sin x < -\frac{1}{2}$ alone.

INEQUALITY	IN t	SURVIVING CONDITION
$2\sin^2x - \sin x - 1 > 0$	$t < -\frac{1}{2}$ or $t > 1$	$\sin x < -\frac{1}{2}$
$\cos^2x \leq \frac{1}{4}$	$-\frac{1}{2} \leq t \leq \frac{1}{2}$	both branches
$4\sin^2x > 3$	$ t > \frac{\sqrt{3}}{2}$	two pairs of arcs

A compound inequality. $-\frac{1}{2} < \cos x < \frac{\sqrt{2}}{2}$ is an intersection of two arcs. Shade

both on one circle and read the overlap — two intervals, symmetric about the horizontal axis.

02 Worked examples

EXAMPLE 1 Solve $\sin x > \frac{\sqrt{2}}{2}$.

① $\sin x = \frac{\sqrt{2}}{2} \Rightarrow x = \frac{\pi}{4}, \frac{3\pi}{4}$

The boundary points.

② Shade the arc above the chord.

It runs anticlockwise from $\frac{\pi}{4}$.

③ $\frac{\pi}{4} + 2\pi n < x < \frac{3\pi}{4} + 2\pi n$

ANSWER

$$\frac{\pi}{4} + 2\pi n < x < \frac{3\pi}{4} + 2\pi n$$

EXAMPLE 2 Solve $\cos x \leq -\frac{1}{2}$.

① $\arccos(-\frac{1}{2}) = \frac{2\pi}{3}$

② The cosine is small on the **left** arc.

③ $\frac{2\pi}{3} + 2\pi n \leq x \leq \frac{4\pi}{3} + 2\pi n$

Closed — the inequality is not strict.

ANSWER

$$\frac{2\pi}{3} + 2\pi n \leq x \leq \frac{4\pi}{3} + 2\pi n$$

EXAMPLE 3 Solve $\tan x \geq 1$.

$$\textcircled{1} \quad \arctan 1 = \frac{\pi}{4}$$

$$\textcircled{2} \quad \text{The tangent increases to } +\infty \text{ at } \frac{\pi}{2}.$$

$$\textcircled{3} \quad \frac{\pi}{4} + \pi n \leq x < \frac{\pi}{2} + \pi n$$

Open at the asymptote.

ANSWER

$$\frac{\pi}{4} + \pi n \leq x < \frac{\pi}{2} + \pi n$$

EXAMPLE 4 Solve $2\sin^2 x - \sin x - 1 > 0$ on $[0, 2\pi]$.

$$\textcircled{1} \quad (2t + 1)(t - 1) > 0$$

With $t = \sin x$.

$$\textcircled{2} \quad t < -\frac{1}{2} \text{ or } t > 1; \text{ the second is empty.}$$

$$\textcircled{3} \quad \sin x < -\frac{1}{2}$$

$$\textcircled{4} \quad \frac{7\pi}{6} < x < \frac{11\pi}{6}$$

ANSWER

$$\frac{7\pi}{6} < x < \frac{11\pi}{6}$$

03 Interactive model

Give each pair a paper circle and a ruler for the chord; they shade, then read the interval aloud.

From the sign of an expression to an interval

$x > 2$



A horizontal number line with tick marks from -8 to 8. The tick marks are labeled -8, -6, -4, -2, 0, 2, 4, 6, 8. An open circle is drawn at the number 2. The region to the right of the open circle is shaded in dark teal, representing the interval x > 2.

inequality $x > 2$

boundary 2 (open)

$x = 0$ works? no

$x = 6$ works? yes

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Inequality	Tengsizlik	Неравенство
Arc	Yoy	Дуга
Interval of solutions	Yechimlar oralig‘i	Интервал решений
Strict inequality	Qat‘iy tengsizlik	Строгое неравенство
Union of intervals	Oraliqlar birlashmasi	Объединение промежутков
Endpoint	Chetki nuqta	Концевая точка
Period	Davr	Период
Sign chart	Ishoralar jadvali	Таблица знаков

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$\sin x > a$ is solved by reading:

two points

the arc between them

the whole circle

nothing

The arc must be read:

clockwise

anticlockwise

either way

from the larger point

$\cos x > \frac{1}{2}$ on one period is:

$(0, \frac{\pi}{3})$

$(-\frac{\pi}{3}, \frac{\pi}{3})$

$(\frac{\pi}{3}, \frac{5\pi}{3})$

$(0, 2\pi)$

A tangent inequality steps by:

$$2\pi$$

$$\pi$$

$$\frac{\pi}{2}$$

$$\pi \text{ or } 2\pi$$

The interval for $\tan x \geq 1$ is:

closed at both ends

open at $\frac{\pi}{2}$

open at both ends

a single point

$\sin x > 2$ has:

one interval

two intervals

no solution

all of $\mathbb{R}$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $\sin x > 0$

ANSWER $2\pi n < x < \pi + 2\pi n$

2. Solve $\cos x > 0$

ANSWER $-\frac{\pi}{2} + 2\pi n < x < \frac{\pi}{2} + 2\pi n$

3. Solve $\sin x < 0$

ANSWER $\pi + 2\pi n < x < 2\pi + 2\pi n$

4. Solve $\tan x > 0$

ANSWER $\pi n < x < \frac{\pi}{2} + \pi n$

5. Solve $\cos x \geq 1$

ANSWER $x = 2\pi n$

6. Has $\sin x \geq 1.2$ a solution?

ANSWER No

7. Solve $\sin x \leq 1$

ANSWER All x

Medium · the standard question

7 PROBLEMS

1. Solve $\sin x > \frac{1}{2}$

ANSWER $\frac{\pi}{6} + 2\pi n < x < \frac{5\pi}{6} + 2\pi n$

2. Solve $\cos x < \frac{1}{2}$

ANSWER $\frac{\pi}{3} + 2\pi n < x < \frac{5\pi}{3} + 2\pi n$

3. Solve $\sin x \geq \frac{\sqrt{3}}{2}$

ANSWER $\frac{\pi}{3} + 2\pi n \leq x \leq \frac{2\pi}{3} + 2\pi n$

4. Solve $\cos x \leq -\frac{\sqrt{2}}{2}$

ANSWER $\frac{3\pi}{4} + 2\pi n \leq x \leq \frac{5\pi}{4} + 2\pi n$

5. Solve $\tan x < \sqrt{3}$

ANSWER $-\frac{\pi}{2} + \pi n < x < \frac{\pi}{3} + \pi n$

6. Solve $\sin x > -\frac{1}{2}$

ANSWER $-\frac{\pi}{6} + 2\pi n < x < \frac{7\pi}{6} + 2\pi n$

7. Length of the arc for $\cos x > -\frac{1}{2}$

ANSWER $\frac{4\pi}{3}$

Hard · stretch and reason

7 PROBLEMS

1. Solve $2\sin^2 x - \sin x - 1 > 0$ on $[0, 2\pi]$

ANSWER $\frac{7\pi}{6} < x < \frac{11\pi}{6}$

2. Solve $4\cos^2 x \leq 3$ on $[0, \pi]$

ANSWER $\frac{\pi}{6} \leq x \leq \frac{5\pi}{6}$

3. Solve $\sin 2x > \frac{1}{2}$ on $[0, \pi]$

ANSWER $\frac{\pi}{12} < x < \frac{5\pi}{12}$

4. Solve $-\frac{1}{2} < \cos x < \frac{\sqrt{2}}{2}$ on $[0, 2\pi]$

ANSWER $(\frac{\pi}{4}, \frac{2\pi}{3}) \cup (\frac{4\pi}{3}, \frac{7\pi}{4})$

5. Solve $\sin x \cos x > 0$ on $[0, 2\pi]$

ANSWER $(0, \frac{\pi}{2}) \cup (\pi, \frac{3\pi}{2})$

6. Solve $\tan^2 x > 1$ on $(-\frac{\pi}{2}, \frac{\pi}{2})$

ANSWER $(-\frac{\pi}{2}, -\frac{\pi}{4}) \cup (\frac{\pi}{4}, \frac{\pi}{2})$

7. Solve $\sin x + \cos x > 1$ on $[0, 2\pi]$

ANSWER $(0, \frac{\pi}{2})$

07 Homework

Homework — 5 tasks

Draw and shade a circle for every question; the picture is half the mark.

1. Solve $\sin x \geq \frac{1}{2}$ and give the answer on $[0, 2\pi]$.
2. Solve $\cos x > -\frac{\sqrt{3}}{2}$, and state the length of the arc.
3. Solve $\tan x \leq -1$, taking care over the asymptote.
4. Solve $2\cos^2 x - \cos x - 1 < 0$ on $[0, 2\pi]$.
5. Explain, with one circle, why $\sin x > -0.9$ has a much longer solution arc than $\sin x > 0.9$.

Control work 7, and work on the mistakes

Trigonometric equations and inequalities in one paper, then a named diagnosis of each slip.

LESSON 89–90

2 × 40 MIN

ALGEBRA 10, NAZORAT ISHI 7

P1 5 REVIEW

LESSON CLOCK

3

42'

12'

25'

8'

Instructions	3 min
The paper	42 min
Answers	12 min
Diagnosis and rewrite	25 min
Targets	8 min

LEARNING OBJECTIVES

- Solve the four basic equations and their standard extensions under time.
- Select solutions in a given interval without losing a family.
- Solve an inequality and write the answer as intervals.
- Name each lost mark and rewrite the solution in full.

TEXTBOOK

National	pp. 301–304
Cambridge	Pure Mathematics 1, pp. 121–124
Workbook	Control paper A7

01

Explanation

The paper — 35 marks, 42 minutes

Q	TASK	MARKS	FROM
1	Solve $\sin x = -\frac{\sqrt{3}}{2}$ in general and on $[0, 2\pi]$	5	L79–81
2	Solve $\cos 2x = \frac{1}{2}$ on $[0^\circ, 360^\circ]$	5	L79–81
3	Solve $2\sin^2x + 3\sin x - 2 = 0$, rejecting the impossible root	6	L82–85
4	Solve $\sin 2x = \sqrt{3}\cos x$ by factorising	6	L82–85
5	Write $\sin x + \sqrt{3}\cos x$ as $R\sin(x + \alpha)$, then solve $= 1$	7	L82–85
6	Solve $\cos x < \frac{1}{2}$ and give the answer on $[0, 2\pi]$	6	L86–88

WHERE THE MARKS ACTUALLY GO

Q1 and Q2 each carry one mark for the **second** family, Q3 one for rejecting $t = -2$, Q4 one for factorising rather than dividing, Q6 one for reading the arc anticlockwise. Five of the thirty-five marks are for not losing something.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
lost family	one answer where two were wanted	draw the chord and count the points
divided by $\cos x$	$\sin 2x = \sqrt{3} \cos x \Rightarrow 2 \sin x = \sqrt{3}$	factor; keep $\cos x = 0$
root not rejected	$\sin x = -2$	check $ t \leq 1$ before solving
wrong period	$\tan x = a \Rightarrow \dots + 2\pi n$	tangent steps by π
arc read backwards	an empty interval	anticlockwise, always
substitution not undone	answer given in t or in $2x$	divide by the coefficient at the end
$\arccos(-a)$ mishandled	$-\arccos a$	$\pi - \arccos a$

Write the slip's name in the margin of every question you lost a mark on, then rewrite the whole solution — not the wrong line.

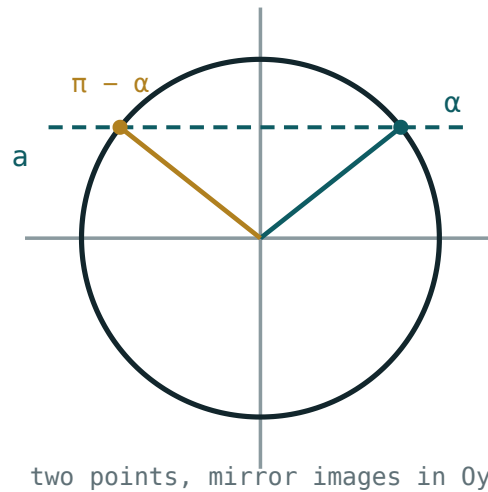
The quarter so far, in one page

EVERYTHING OF LESSONS 79–88

$$\sin x = a \Rightarrow x = (-1)^n \arcsin a + \pi n \quad \cos x = a \Rightarrow x = \pm \arccos a + 2\pi n$$

$$\tan x = a \Rightarrow x = \arctan a + \pi n \quad a \sin x + b \cos x = R \sin(x + \alpha), R = \sqrt{a^2 + b^2}$$

Three moves for the rest: **substitute**, **factorise**, **reduce**. Inequalities: solve the equation, shade the arc, read it anticlockwise.



One picture that generates every formula above.

Lessons 91–98 leave trigonometry entirely for probability. Nothing of this block is needed there — but all of it returns in Grade 11, where the derivative of $\sin x$ makes these equations the last step of an optimisation rather than the whole question.

02 Worked examples

EXAMPLE 1 Model answer, Q2: solve $\cos 2x = \frac{1}{2}$ on $[0^\circ, 360^\circ]$.

- ① Let $t = 2x$, so $t \in [0^\circ, 720^\circ]$.
Widen the interval first.

- ② $t = \pm 60^\circ + 360^\circ n \Rightarrow t = 60^\circ, 300^\circ, 420^\circ, 660^\circ$
Four values in range.

- ③ $x = 30^\circ, 150^\circ, 210^\circ, 330^\circ$
Halve each.

ANSWER

$30^\circ, 150^\circ, 210^\circ, 330^\circ$

EXAMPLE 2 Model answer, Q4: solve $\sin 2x = \sqrt{3} \cos x$.

$$(1) \quad 2 \sin x \cos x - \sqrt{3} \cos x = 0$$

$$(2) \quad \cos x (2 \sin x - \sqrt{3}) = 0$$

Factor, do not divide.

$$(3) \quad x = \frac{\pi}{2} + \pi n$$

$$(4) \quad \sin x = \frac{\sqrt{3}}{2} \Rightarrow x = (-1)^n \frac{\pi}{3} + \pi n$$

ANSWER

$$\frac{\pi}{2} + \pi n \text{ and } (-1)^n \frac{\pi}{3} + \pi n$$

EXAMPLE 3 Model answer, Q5: write $\sin x + \sqrt{3} \cos x$ as $R \sin(x + \alpha)$ and solve $= 1$.

$$(1) \quad R = \sqrt{1+3} = 2, \tan \alpha = \sqrt{3} \Rightarrow \alpha = \frac{\pi}{3}$$

$$(2) \quad 2 \sin(x + \frac{\pi}{3}) = 1$$

$$(3) \quad x + \frac{\pi}{3} = \frac{\pi}{6} + 2\pi n \text{ or } \frac{5\pi}{6} + 2\pi n$$

$$(4) \quad x = -\frac{\pi}{6} + 2\pi n \text{ or } x = \frac{\pi}{2} + 2\pi n$$

ANSWER

$$x = -\frac{\pi}{6} + 2\pi n, x = \frac{\pi}{2} + 2\pi n$$

03 Interactive model

Work Q4 twice on the board — once by factorising, once by dividing — and let the class name the lost family.

Ten questions on the trigonometry block

$\sin x = 1.5$ has:

two solutions

one

none

infinitely many

$\cos x = a$ steps by:

π

2π

$\frac{\pi}{2}$

4π

$\tan x = a$ steps by:

π

2π

$\frac{\pi}{2}$

3π

$\arccos(-\frac{\sqrt{2}}{2})$ is:

$$-\frac{\pi}{4}$$

$$\frac{3\pi}{4}$$

$$\frac{\pi}{4}$$

$$\frac{5\pi}{4}$$

To solve $2\cos^2x + \sin x = 2$ you first:

divide by $\cos x$

replace $\cos^2x$ by $1 - \sin^2x$

square

substitute $t = 2x$

$\sin x \cos x = \sin x$ should be:

divided

factorised

squared

left alone

A homogeneous equation is divided by:

$\sin x$

a power of $\cos x$

2

nothing

The greatest value of $5 \sin x + 12 \cos x$ is:

12

13

17

5

$\sin x > a$ gives:

points

arcs

one number

nothing

$\tan x \geq 1$ is open at:

$$\frac{\pi}{4}$$

$$\frac{\pi}{2}$$

both ends

neither end

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
General solution	Umumiy yechim	Общее решение
Family of solutions	Yechimlar oilasi	Семейство решений
Homogeneous	Bir jinsli	Однородное
Auxiliary angle	Yordamchi burchak	Вспомогательный угол
Extraneous root	Chet ildiz	Посторонний корень
Diagnosis	Tashxis	Диагностика
Target	Maqsad	Цель

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The most common slip in Q1 is:

arithmetic

losing the second family

the period

the sign of π

Q3 requires you to:

reject a root outside $[-1, 1]$

square both sides

divide by $\sin x$

use R

After substituting $t = 2x$ you must:

stop

halve the answers

double them

add π

Lessons 91–98 turn to:

logarithms

probability

vectors

integration

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $\sin x = -\frac{\sqrt{3}}{2}$

ANSWER $x = (-1)^{n+1} \frac{\pi}{3} + \pi n$

2. Solve $\cos x = \frac{1}{2}$

ANSWER $x = \pm \frac{\pi}{3} + 2\pi n$

3. Solve $\tan x = -1$

ANSWER $x = -\frac{\pi}{4} + \pi n$

4. R for $\sin x + \sqrt{3} \cos x$

ANSWER 2

5. Reject or keep $t = -2$ in $\sin x = t$

ANSWER Reject

6. Solve $\cos x < \frac{1}{2}$ on $[0, 2\pi]$

ANSWER $(\frac{\pi}{3}, \frac{5\pi}{3})$

7. Period of the tangent

ANSWER π

Medium · the standard question

7 PROBLEMS

1. Solve $\cos 2x = \frac{1}{2}$ on $[0^\circ, 360^\circ]$

ANSWER $30^\circ, 150^\circ, 210^\circ, 330^\circ$

2. Solve $2\sin^2 x + 3 \sin x - 2 = 0$

ANSWER $x = (-1)^n \frac{\pi}{6} + \pi n$

3. Solve $\sin 2x = \sqrt{3} \cos x$

ANSWER $\frac{\pi}{2} + \pi n; (-1)^n \frac{\pi}{3} + \pi n$

4. Solve $\sin x + \sqrt{3} \cos x = 1$

ANSWER $-\frac{\pi}{6} + 2\pi n; \frac{\pi}{2} + 2\pi n$

5. Solve $\sin x \geq \frac{1}{2}$ on $[0, 2\pi]$

ANSWER $[\frac{\pi}{6}, \frac{5\pi}{6}]$

6. Solve $\tan 2x = 1$ on $[0, \pi]$

ANSWER $\frac{\pi}{8}, \frac{5\pi}{8}$

7. Greatest value of $\sin x + \sqrt{3} \cos x$

ANSWER 2

Hard · stretch and reason

7 PROBLEMS

1. Solve $\cos 2x + 3 \sin x = 2$ on $[0, 2\pi]$

ANSWER $\frac{\pi}{6}, \frac{5\pi}{6}$

2. Solve $3\sin^2 x - 4 \sin x \cos x + \cos^2 x = 0$

ANSWER $\frac{\pi}{4} + \pi n; \arctan \frac{1}{3} + \pi n$

3. Solve $\sin 3x = \sin x$ on $[0, \pi]$

ANSWER $0, \frac{\pi}{4}, \frac{3\pi}{4}, \pi$

4. Solve $|\sin x| > \frac{1}{2}$ on $[0, 2\pi]$

ANSWER $(\frac{\pi}{6}, \frac{5\pi}{6}) \cup (\frac{7\pi}{6}, \frac{11\pi}{6})$

5. For which a has $2\sin^2 x - \sin x + a = 0$ a solution?

ANSWER $-3 \leq a \leq \frac{1}{8}$

6. Solve $\sin x + \cos x = \sqrt{2}$ on $[0, 2\pi]$

ANSWER $\frac{\pi}{4}$

7. Solve $\cos^2 x - \sin^2 x > \frac{1}{2}$ on $[0, 2\pi]$

ANSWER $(0, \frac{\pi}{6}) \cup (\frac{5\pi}{6}, \frac{7\pi}{6}) \cup (\frac{11\pi}{6}, 2\pi)$

97 Homework

Homework — 4 tasks

Bring the one-page summary to Lesson 91; probability begins from a blank page.

1. Rewrite in full every control-work question that lost a mark, naming the slip in the margin.

2. Write the four general solutions and the three moves on one page, with a circle drawn for each.
3. Solve $2\cos^2 x - 5 \cos x + 2 = 0$ on $[0, 2\pi]$.
4. Write your target for the probability block in one checkable sentence, and date it.

Random events

The sample space first, the event second, the probability only third.

LESSON 91–93

3 × 40 MIN

ALGEBRA 10, §29

IGX 24.1

LESSON CLOCK

15'

30'

30'

30'

25'

5

Three kinds of event	15 min
The sample space	30 min
Tables and trees	30 min
Combining events	30 min
Exclusive and independent	25 min
Homework	5 min

LEARNING OBJECTIVES

- Distinguish certain, impossible and random events.
- List a sample space systematically, by table or by tree.
- Describe the union, intersection and complement of events.
- Recognise when two events are mutually exclusive or independent.

TEXTBOOK

National	pp. 305–318
Cambridge	Core & Extended, pp. 620–628
Workbook	Exercise 24.1

01

Explanation

Three kinds of event

A **trial** is anything with an uncertain result: a die thrown, a card drawn, a bulb tested. Its possible results are **outcomes**; a set of outcomes is an **event**.

KIND	HAPPENS	EXAMPLE (ONE DIE)
certain	always	“the score is under 7”
impossible	never	“the score is 8”
random	sometimes	“the score is even”

NOTATION USED THROUGHOUT

Events are $A, B, C, \dots$; the certain event is U (or Ω); the impossible event is $\emptyset$. The complement of A — “not A ” — is written A' or $\bar{A}$.

The sample space

Every probability question begins the same way: **list what can happen**. The list is the sample space, written U , and everything else is counted inside it.

TRIAL	SAMPLE SPACE	$ U $
one coin	$\{H, T\}$	2
one die	$\{1, 2, 3, 4, 5, 6\}$	6
two coins	$\{HH, HT, TH, TT\}$	4
two dice	ordered pairs (a, b)	36
a card from 52	the 52 cards	52
three coins	$\{HHH, \dots, TTT\}$	8

HT AND TH ARE DIFFERENT OUTCOMES

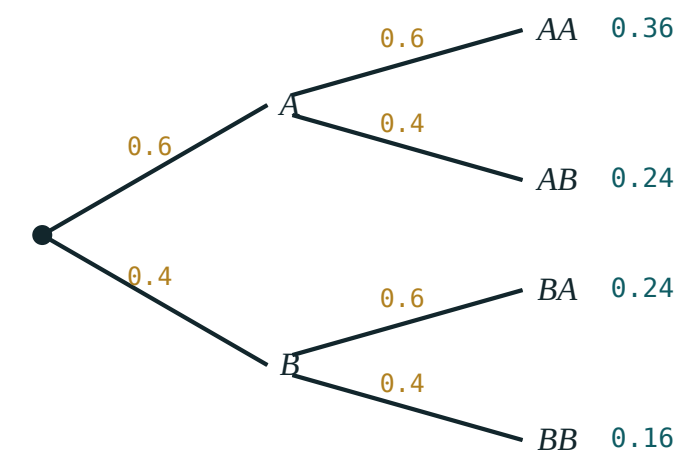
With two coins there are four equally likely outcomes, not three. “One head and one tail” happens in **two** of them, which is why its probability is $\frac{2}{4}$ and not $\frac{1}{3}$. Order the outcomes and the trap disappears.

Tables and trees

Two devices make the list systematic.

A table suits two trials with few outcomes each — two dice, for instance, give a 6×6 grid whose 36 cells are the sample space. Reading “the total is 7” off the leading anti-diagonal gives 6 cells at once.

A tree suits a sequence of stages, especially when the stages differ or when something is removed between them.



multiply along a branch · add down the ends

Each path through the tree is one outcome; the paths together are the sample space.

TWO RULES FOR TREES

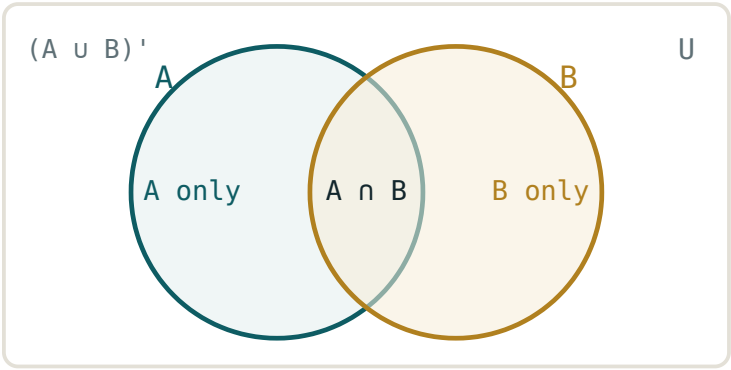
- **Along** a branch, multiply. The probabilities of one path multiply together.
- **Down** the ends, add. Several paths that all count as the event are added.

The probabilities on any one fork must total 1 — a check worth doing every time.

Combining events

Events are sets, so the set operations describe them.

IN WORDS	NOTATION	MEANS
A or B (or both)	$A \cup B$	the union
A and B	$A \cap B$	the intersection
not A	A'	the complement
A but not B	$A \cap B'$	the difference



$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Two events, four regions — and every question in this block names one of them.

One die. Let $A = \text{“even”} = \{2, 4, 6\}$ and $B = \text{“greater than 3”} = \{4, 5, 6\}$. Then $A \cup B = \{2, 4, 5, 6\}$, $A \cap B = \{4, 6\}$, $A' = \{1, 3, 5\}$ and $A \cap B' = \{2\}$.

Exclusive and independent

Two words are constantly confused; they describe entirely different things.

	MUTUALLY EXCLUSIVE	INDEPENDENT
meaning	cannot both happen	one does not affect the other
test	$A \cap B = \emptyset$	$P(A \cap B) = P(A) \cdot P(B)$
Venn picture	circles apart	circles overlap, in proportion
example	“even” and “odd”	two throws of one die

EXCLUSIVE EVENTS ARE NEVER INDEPENDENT

If A and B cannot both happen, then knowing A occurred tells you B did **not** — that is the strongest possible dependence. The two words are almost opposites, not synonyms.

Replacement decides independence. Drawing two balls *with* replacement leaves the second draw untouched; drawing *without* replacement changes it. The phrase “without replacement” in a question is an instruction to change the second row of the tree.

02 Worked examples

EXAMPLE 1 Two dice are thrown. List the sample space and find how many outcomes give a total of 7.

- ① Ordered pairs (a, b) with $a, b \in \{1, \dots, 6\}$.
 $6 \times 6 = 36$ outcomes.

- ② $(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)$

- ③ Six favourable outcomes.

ANSWER

$|U| = 36$; six outcomes give 7

EXAMPLE 2 A die is thrown. With $A = \text{“even”}$ and $B = \text{“prime”}$, list $A \cap B$ and $A \cup B$.

- ① $A = \{2, 4, 6\}$

- ② $B = \{2, 3, 5\}$
 1 is not prime.

- ③ $A \cap B = \{2\}$

- ④ $A \cup B = \{2, 3, 4, 5, 6\}$

ANSWER

$A \cap B = \{2\}, A \cup B = \{2, 3, 4, 5, 6\}$

EXAMPLE 3 A bag holds 3 red and 2 blue balls. Two are drawn without replacement. Draw the tree and list the outcomes.

① First fork: R with $\frac{3}{5}$, B with $\frac{2}{5}$.

② After R : $\frac{2}{4}$ and $\frac{2}{4}$.

Four balls left.

③ After B : $\frac{3}{4}$ and $\frac{1}{4}$.

④ Outcomes RR , RB , BR , BB .

ANSWER

Four outcomes, with the second fork depending on the first

EXAMPLE 4 Are “the first throw is a 6” and “the second throw is a 6” exclusive, independent, or neither?

① They can both happen — the throw $(6, 6)$.

Not exclusive.

② The first throw does not change the die.

③ $P(A \cap B) = \frac{1}{36} = \frac{1}{6} \times \frac{1}{6}$

Independent.

ANSWER

Independent, and not mutually exclusive

03 Interactive model

Throw two dice thirty times and tally the totals on the board; the shape of the answer appears before any formula.

Naming events correctly

The sample space for two coins has:

“The score on a die is 8” is:

A' means:

Two dice give how many outcomes?

12

21

36

6

Exclusive events satisfy:

$$A \cap B = \emptyset$$

$$A \cup B = U$$

$$P(A) = P(B)$$

$$A = B$$

Along a branch of a tree you:

add

multiply

subtract

divide

Drawing without replacement makes the draws:

independent

dependent

exclusive

certain

The probabilities on one fork total:

0

1

2

anything

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Random event	Tasodifiy hodisa	Случайное событие
Trial (experiment)	Sinov	Испытание
Outcome	Natija	Исход
Sample space	Elementar hodisalar fazosi	Пространство элементарных исходов
Certain event	Muqarrar hodisa	Достоверное событие
Impossible event	Mumkin bo'lmagan hodisa	Невозможное событие
Complementary event	Qarama-qarshi hodisa	Противоположное событие
Mutually exclusive	Birgalikda bo'lmagan	Несовместные
Independent events	Bog'liqsiz hodisalar	Независимые события
Tree diagram	Daraxt diagrammasi	Дерево исходов
Equally likely	Teng imkoniyatli	Равновозможные
Favourable outcome	Qulay natija	Благоприятный исход

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A sample space lists:

the favourable outcomes

every possible outcome

the events

the probabilities

With two coins, “one head” has:

one outcome

two outcomes

three

four

Mutually exclusive events are:

always independent

never independent (unless one is impossible)

the same thing

complementary

A tree diagram suits:

a single throw

a sequence of stages

a Venn question

a certain event

$A \cap B'$ means:

A or B

A but not B

neither

both

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Sample space for one coin

ANSWER $\{H, T\}$

2. Number of outcomes for two coins

ANSWER 4

3. Number of outcomes for two dice

ANSWER 36

4. Is “a die shows 7” certain, impossible or random?

ANSWER Impossible

5. Is “a die shows less than 7” certain, impossible or random?

ANSWER Certain

6. A = even on a die: list A'

ANSWER $\{1, 3, 5\}$

7. Number of outcomes for three coins

ANSWER 8

Medium · the standard question

7 PROBLEMS

1. $A = \text{even}, B = \text{prime on a die}: A \cap B$

ANSWER $\{2\}$

2. Same: $A \cup B$

ANSWER $\{2, 3, 4, 5, 6\}$

3. Two dice: how many outcomes give a total of 8?

ANSWER 5

4. Two dice: how many give a double?

ANSWER 6

5. Two coins: outcomes with exactly one head

ANSWER 2

6. A bag of 3 red, 2 blue: the second fork after a red, without replacement

ANSWER $\frac{2}{4}$ red, $\frac{2}{4}$ blue

7. Are “even” and “odd” on one die exclusive?

ANSWER Yes

Hard · stretch and reason

7 PROBLEMS

1. Two dice: how many outcomes have a total that is prime?

ANSWER 15

2. Three coins: how many outcomes have at least two heads?

ANSWER 4

3. A card from 52: are “red” and “king” exclusive? independent?

ANSWER Not exclusive; independent

4. Two dice: describe $A \cap B$ where A = “first is even”, B = “total is 5”

ANSWER $\{(2,3), (4,1)\}$

5. A bag of 4 white and 6 black: all outcomes of two draws without replacement

ANSWER WW, WB, BW, BB with changing forks

6. Explain why exclusive events with $P(A), P(B) > 0$ cannot be independent

ANSWER $P(A \cap B) = 0 \neq P(A)P(B)$

7. Two dice: how many outcomes give a total of at least 10?

ANSWER 6

07 Homework

Homework — 5 tasks

Every answer starts with the sample space written out or counted explicitly.

- List the sample space for throwing a coin and a die together, and say how many outcomes it has.
- For one die, let A = “multiple of 3” and B = “greater than 2”. List $A \cup B$, $A \cap B$, A' and $A \cap B'$.
- Draw the tree for two draws from a bag of 5 green and 3 yellow counters, without replacement.
- Give one pair of events that is exclusive but not independent, and one pair that is independent but not exclusive.

5. Make a 6×6 table for two dice and shade the cells where the total is 9.

Definitions of probability

Classical, statistical and geometric — three definitions, one number between 0 and 1.

LESSON 94–96

3 × 40 MIN

ALGEBRA 10, §30

IGX 24.2–24.4

LESSON CLOCK

25'

25'

25'

35'

20'

5

The classical definition	25 min
Relative frequency	25 min
Geometric probability	25 min
The two rules	35 min
Conditional probability	20 min
Homework	5 min

LEARNING OBJECTIVES

- Use the classical definition $P(A) = m/n$ for equally likely outcomes.
- Estimate a probability from relative frequency, and say when that is necessary.
- Use the addition and multiplication rules, and the complement.
- Compute a simple conditional probability from a table or a tree.

TEXTBOOK

National	pp. 319–334
Cambridge	Core & Extended, pp. 629–642
Workbook	Exercise 24.2–24.4

01

Explanation

The classical definition

When every outcome of a trial is equally likely, probability is a count divided by a count:

$$P(A) = \frac{m}{n} = \frac{\text{favourable outcomes}}{\text{all outcomes}}$$

Every probability lies between 0 and 1, with $P(\varnothing) = 0$ and $P(U) = 1$.

EVENT	M	N	P
a die shows 3	1	6	$\frac{1}{6}$
a die shows an even number	3	6	$\frac{1}{2}$
two dice total 7	6	36	$\frac{1}{6}$
a card is a heart	13	52	$\frac{1}{4}$
two coins give two heads	1	4	$\frac{1}{4}$

“EQUALLY LIKELY” IS A CONDITION, NOT A DECORATION

A drawing pin lands point-up or point-down, but not with probability $\frac{1}{2}$. The classical definition applies only when symmetry guarantees equal outcomes; otherwise you must measure.

Relative frequency

When outcomes are not equally likely — a bent coin, a bus arriving late, a component failing — probability is **estimated** from experiment:

$$P(A) \approx \frac{\text{number of times A happened}}{\text{number of trials}}$$

This is the **statistical** definition. The estimate improves as the number of trials grows, and settles down towards a fixed number — the law of large numbers, seen but not proved here.

THROWS	HEADS	RELATIVE FREQUENCY
10	7	0.700
100	54	0.540
1000	509	0.509
10 000	5023	0.5023

Expected number. If $P(A) = p$ and the trial is repeated N times, the expected number of successes is Np . Rolling a die 300 times, expect $300 \times \frac{1}{6} = 50$ sixes — expect, not guarantee.

Geometric probability

When the outcomes form a continuous region, counting is replaced by measuring:

$$P(A) = \frac{\text{measure of the favourable region}}{\text{measure of the whole region}}$$

The measure is a length, an area or a volume, whichever the problem lives in.

PROBLEM	MEASURE	P
a point on $[0, 10]$ falls in $[3, 5]$	length	0.2
a dart in a square of side 4 hits the inscribed circle	area	$\frac{\pi}{4} \approx 0.785$
a bus is due any time in a 20-minute window; you wait 5	length	0.25

A SINGLE POINT HAS PROBABILITY ZERO

Landing at exactly $x = 3$ has probability 0 — yet it is not impossible. On a continuous space, probability zero and impossible are different things.

The two rules

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Subtract the overlap, or it is counted twice. When A and B are mutually exclusive the overlap is empty and the rule shortens to $P(A) + P(B)$.

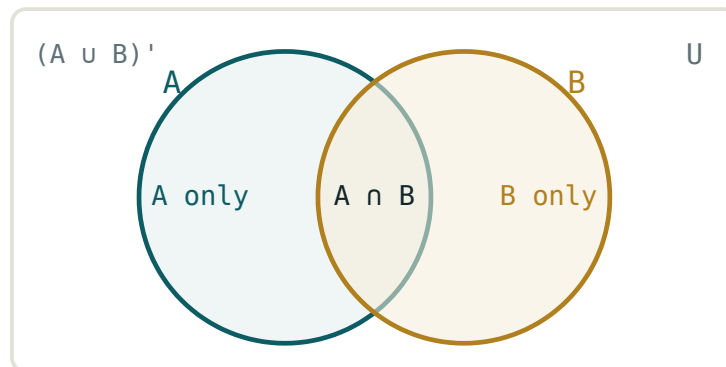
$$P(A \cap B) = P(A) \cdot P(B) \text{ for independent events}$$

$$P(A') = 1 - P(A)$$

THE COMPLEMENT IS THE SHORTEST ROUTE

“At least one six in four throws” has many favourable cases; “no six in four throws” has one description.

So $P = 1 - \left(\frac{5}{6}\right)^4 \approx 0.518$. Whenever a question says **at least**, try the complement first.



$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The overlap subtracted once — the whole of the addition rule in one picture.

Conditional probability

$P(B | A)$ is the probability of B **given that** A has happened. Conditioning shrinks the sample space to A :

$$P(B | A) = \frac{P(A \cap B)}{P(A)}$$

From a table. Of 30 students, 18 study French, 12 of whom also study German.

Then $P(\text{German} | \text{French}) = \frac{12}{18} = \frac{2}{3}$ — the denominator is 18, not 30. That single change of denominator is the whole idea.

From a tree. The numbers on the second row of forks are already conditional probabilities. A bag of 3 red and 2 blue, drawn without replacement: $P(\text{second red} | \text{first red}) = \frac{2}{4}$.

$P(B | A)$ AND $P(A | B)$ ARE DIFFERENT NUMBERS

Most people who are ill have a temperature; most people with a temperature are not seriously ill. Swapping the two is the commonest error in the whole of probability.

02 Worked examples

EXAMPLE 1 Two dice are thrown. Find the probability that the total is 7 or 11.

- ① Total 7: 6 outcomes; total 11: 2.

Exclusive events.

②
$$P = \frac{6}{36} + \frac{2}{36}$$

③
$$= \frac{8}{36} = \frac{2}{9} \approx 0.222$$

ANSWER

$$\frac{2}{9}$$

EXAMPLE 2 A coin is thrown four times. Find the probability of at least one head.

- ① Complement: no head at all.

②
$$P(\text{no head}) = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$$

③
$$P(\text{at least one}) = 1 - \frac{1}{16} = \frac{15}{16}$$

ANSWER

$$\frac{15}{16} = 0.9375$$

EXAMPLE 3 A bag holds 3 red and 2 blue balls. Two are drawn without replacement. Find $P(\text{both red})$ and $P(\text{one of each})$.

$$\textcircled{1} \quad P(RR) = \frac{3}{5} \times \frac{2}{4} = \frac{3}{10}$$

Multiply along the branch.

$$\textcircled{2} \quad P(RB) = \frac{3}{5} \times \frac{2}{4} = \frac{3}{10}$$

$$\textcircled{3} \quad P(BR) = \frac{2}{5} \times \frac{3}{4} = \frac{3}{10}$$

$$\textcircled{4} \quad P(\text{one of each}) = \frac{3}{5}$$

Add down the ends.

ANSWER

$$\frac{3}{10} \text{ and } \frac{3}{5}$$

EXAMPLE 4 Of 40 pupils, 25 play football, 18 play chess and 10 play both. Find $P(\text{football} \mid \text{chess})$.

$$\textcircled{1} \quad P(F \cap C) = \frac{10}{40}$$

$$\textcircled{2} \quad P(C) = \frac{18}{40}$$

$$\textcircled{3} \quad P(F \mid C) = \frac{10}{18} = \frac{5}{9}$$

The denominator shrinks to 18.

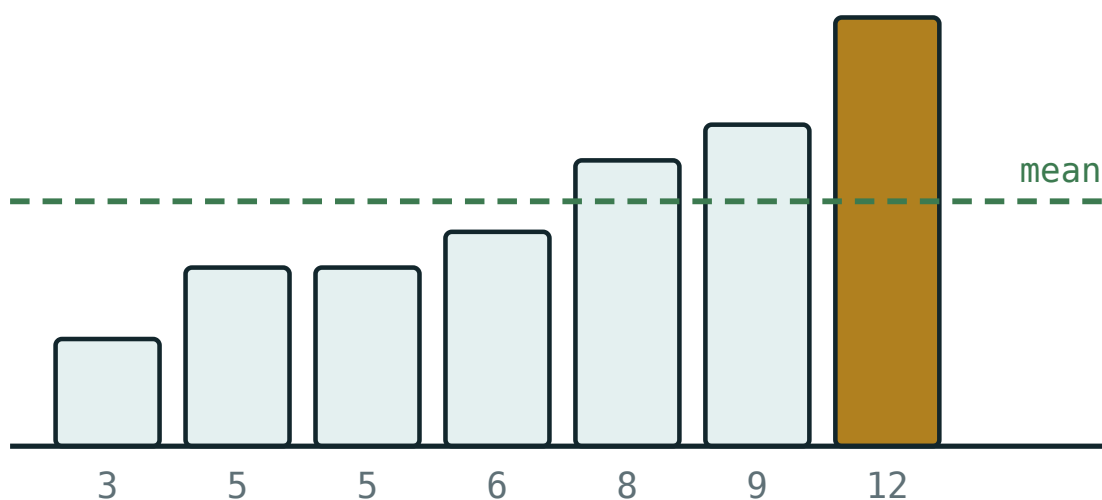
ANSWER

$$\frac{5}{9} \approx 0.556$$

03 Interactive model

Throw one coin 100 times as a class and plot the running relative frequency — the curve settling on 0.5 is the lesson.

Frequency, and what it settles on



values 3, 5, 5, 6, 8, 9, 12

mean 6.86

median 6

mode 5

range 9

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Probability	Ehtimollik	Вероятность
Classical definition	Klassik ta'rif	Классическое определение
Relative frequency	Nisbiy chastota	Относительная частота
Statistical probability	Statistik ehtimollik	Статистическая вероятность
Geometric probability	Geometrik ehtimollik	Геометрическая вероятность
Addition rule	Qo'shish qoidasi	Правило сложения
Multiplication rule	Ko'paytirish qoidasi	Правило умножения
Conditional probability	Shartli ehtimollik	Условная вероятность
Complement	To'ldiruvchi	Дополнение
Expected number	Kutilayotgan son	Ожидаемое число

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The classical definition needs the outcomes to be:

few

equally likely

independent

exclusive

$P(A')$ equals:

$1 - P(A)$

$P(A)$

$1 + P(A)$

0

For exclusive events, $P(A \cup B)$ is:

$P(A)P(B)$

$P(A) + P(B)$

$P(A) - P(B)$

1

For independent events, $P(A \cap B)$ is:

“At least one” is usually done by:

$P(B | A)$ has denominator:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. P a die shows 5

ANSWER $\frac{1}{6}$

2. P a die shows an odd number

ANSWER $\frac{1}{2}$

3. P a card is a heart

ANSWER $\frac{1}{4}$

4. P two coins give two tails

ANSWER $\frac{1}{4}$

5. $P(A')$ when $P(A) = 0.3$

ANSWER 0.7

6. Expected sixes in 300 throws of a die

ANSWER 50

7. P a point of $[0, 10]$ lies in $[3, 5]$

ANSWER 0.2

Medium · the standard question

7 PROBLEMS

1. P two dice total 7

ANSWER $\frac{1}{6}$

2. P two dice total 7 or 11

ANSWER $\frac{2}{9}$

3. P at least one head in three throws

ANSWER $\frac{7}{8}$

4. $P(A \cup B)$ when $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.2$

ANSWER 0.7

5. Bag of 3 red, 2 blue, two drawn without replacement: $P(RR)$

ANSWER $\frac{3}{10}$

6. Same bag: P one of each colour

ANSWER $\frac{3}{5}$

7. A dart in a square of side 4 hits the inscribed circle

ANSWER $\frac{\pi}{4} \approx 0.785$

Hard · stretch and reason

7 PROBLEMS

1. P at least one six in four throws of a die

ANSWER $1 - \left(\frac{5}{6}\right)^4 \approx 0.518$

2. Of 40 pupils, 25 play football, 18 chess, 10 both: $P(F | C)$

ANSWER $\frac{5}{9}$

3. Same data: P a pupil plays neither

ANSWER $\frac{7}{40}$

4. Two cards from 52 without replacement: P both aces

ANSWER $\frac{1}{221}$

5. Three coins: P exactly two heads

ANSWER $\frac{3}{8}$

6. A bus is equally likely at any time in 20 minutes; you arrive and wait at most 5: P

ANSWER 0.25

7. If $P(A) = 0.6$, $P(B) = 0.5$ and they are independent, find $P(A \cup B)$

ANSWER 0.8

07 Homework

Homework — 5 tasks

State which definition you are using — classical, statistical or geometric — in every answer.

- Two dice are thrown. Find the probability that the total is 9, and that the total is at least 10.
- A coin is thrown five times. Find the probability of at least one tail.

3. A bag holds 4 white and 6 black balls; two are drawn without replacement. Find the probability that they are the same colour.
4. In a class of 30, 20 study English, 14 study Russian and 8 study both. Find $P(\text{English} \mid \text{Russian})$ and $P(\text{Russian} \mid \text{English})$, and comment on the difference.
5. A point is chosen at random inside a square of side 6. Find the probability that it is within 1 unit of an edge.

Control work 8, and work on the mistakes

The probability block in one paper, then the two rules drawn as a single decision tree.

LESSON 97–98

2 × 40 MIN

ALGEBRA 10, NAZORAT ISHI 8

IGX 24 REVIEW

LESSON CLOCK

3

42'

12'

25'

8'

Instructions	3 min
The paper	42 min
Answers	12 min
Diagnosis and rewrite	25 min
The decision tree	8 min

LEARNING OBJECTIVES

- Compute probabilities from a sample space, a table and a tree under time.
- Choose between the addition rule, the multiplication rule and the complement.
- Compute a conditional probability with the right denominator.
- Name each lost mark and rewrite the solution in full.

TEXTBOOK

National	pp. 335–338
Cambridge	Core & Extended, pp. 643–646
Workbook	Control paper A8

01

Explanation

The paper — 30 marks, 42 minutes

Q	TASK	MARKS	FROM
1	Two dice: find $P(\text{total } 8)$, $P(\text{a double})$, $P(\text{total } 8 \text{ or a double})$	6	L91–96
2	A coin thrown 5 times: find $P(\text{at least one head})$ and $P(\text{exactly one head})$	5	L94–96
3	A bag of 5 red and 4 green, two drawn without replacement: the tree, then $P(\text{same colour})$	6	L91–96
4	Of 50 pupils, 32 study physics, 21 chemistry, 15 both. Find $P(\text{neither})$ and $P(\text{physics} \mid \text{chemistry})$	7	L94–96
5	A point in a square of side 8 lands inside the inscribed circle: find P	3	L94–96
6	State whether two given events are exclusive, independent, or neither, with a reason	3	L91–93

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for subtracting the overlap, Q2 one for using the complement, Q3 one for changing the second fork, Q4 one for the denominator 21 rather than 50. Four of the thirty marks are for a single decision each.

The decision tree for the whole block

THE QUESTION SAYS	USE	FORMULA
“or”, “either”	the addition rule	$P(A) + P(B) - P(A \cap B)$
“and”, “both”, “then”	the multiplication rule	$P(A) \cdot P(B \mid A)$
“at least one”	the complement	$1 - P(\text{none})$
“given that”, “of those who”	conditional	$\frac{P(A \cap B)}{P(B)}$
“without replacement”	a tree with changing forks	second row differs
“at random inside”	geometric probability	$\frac{\text{area}}{\text{area}}$

THE FOUR SLIPS THAT ACCOUNT FOR MOST LOST MARKS

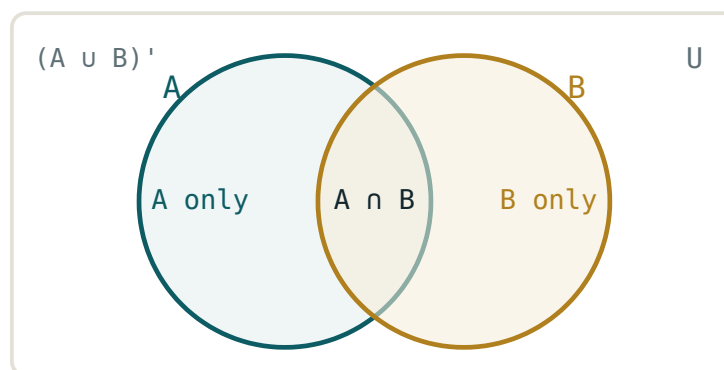
- adding without subtracting the overlap;
- multiplying dependent events as though they were independent;
- using $|U|$ as the denominator of a conditional probability;
- forgetting that “without replacement” changes the second fork.

Looking forward

Lessons 99–102 are the Cambridge revision block: sequences, arithmetic and geometric progressions, infinite geometric series and the binomial expansion. None of it is new to the national syllabus of Grade 9, but the notation is Cambridge's and the questions are harder.

ONE HABIT TO CARRY FORWARD

Probability rewarded writing the sample space before anything else. Series reward writing a and d (or r) before anything else. In both, one line of setting-up prevents three lines of error.



$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The picture behind Q1 and Q4 alike.

02 Worked examples

EXAMPLE 1 Model answer, Q1: two dice, $P(\text{total } 8 \text{ or a double})$.

① Total 8: $(2,6),(3,5),(4,4),(5,3),(6,2) — 5$.

② Doubles: 6.

③ Both: $(4,4) — 1$.

The overlap.

④
$$P = \frac{5}{36} + \frac{6}{36} - \frac{1}{36} = \frac{10}{36} = \frac{5}{18}$$

ANSWER

$$\frac{5}{18} \approx 0.278$$

EXAMPLE 2 Model answer, Q3: 5 red and 4 green, two drawn without replacement, $P(\text{same colour})$.

①
$$P(RR) = \frac{5}{9} \times \frac{4}{8} = \frac{5}{18}$$

②
$$P(GG) = \frac{4}{9} \times \frac{3}{8} = \frac{1}{6}$$

③
$$P = \frac{5}{18} + \frac{3}{18} = \frac{8}{18} = \frac{4}{9}$$

Add down the ends.

ANSWER

$$\frac{4}{9} \approx 0.444$$

EXAMPLE 3 Model answer, Q4: 50 pupils, 32 physics, 21 chemistry, 15 both.

$$① \quad |P \cup C| = 32 + 21 - 15 = 38$$

$$② \quad P(\text{neither}) = \frac{12}{50} = 0.24$$

$$③ \quad P(\text{physics} \mid \text{chemistry}) = \frac{15}{21} = \frac{5}{7}$$

Denominator 21.

ANSWER

0.24 and $\frac{5}{7}$

03 Interactive model

Work Q4 twice — once with denominator 50, once with 21 — and let the class say which question each answers.

Ten questions on the probability block

Two dice give how many outcomes?

12

36

21

6

$P(A')$ is:

$1 - P(A)$

$P(A)$

0

$1 + P(A)$

The addition rule subtracts:

nothing

the overlap

the smaller probability

1

Independent events multiply because:

they overlap

one does not affect the other

they are exclusive

of the tree

“At least one” suggests:

a long list

the complement

geometry

conditioning

$P(B | A)$ divides by:

$P(B)$

$P(A)$

1

$|U|$

“Without replacement” changes:

nothing

the second fork

the sample space size only

the first fork

Geometric probability divides:

counts

measures

frequencies

nothing

Expected sixes in 600 throws:

60

100

120

600

Exclusive events with positive probability are:

independent

not independent

certain

complementary

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Sample space	Elementar hodisalar fazosi	Пространство исходов
Addition rule	Qo'shish qoidasi	Правило сложения
Multiplication rule	Ko'paytirish qoidasi	Правило умножения
Complement	To'ldiruvchi	Дополнение
Conditional probability	Shartli ehtimollik	Условная вероятность
Without replacement	Qaytarmasdan	Без возвращения
Target	Maqsad	Цель

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The overlap in Q1 is:

(4,4)

(2,6)

none

all doubles

Q3 needs a tree because:

there are two colours

the second draw depends on the first

it is longer

no reason

The denominator in Q4's conditional part is:

50

32

21

15

Lessons 99–102 revise:

trigonometry

progressions and the binomial expansion

geometry

logarithms

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. P two dice total 8

ANSWER $\frac{5}{36}$

2. P two dice show a double

ANSWER $\frac{1}{6}$

3. P at least one head in five throws

ANSWER $\frac{31}{32}$

4. P exactly one head in five throws

ANSWER $\frac{5}{32}$

5. A point in a square of side 8 hits the inscribed circle

ANSWER $\frac{\pi}{4} \approx 0.785$

6. $P(A \cup B)$ when $P(A) = 0.4$, $P(B) = 0.3$ and exclusive

ANSWER 0.7

7. Are “even” and “multiple of 3” on a die exclusive?

ANSWER No — 6 is both

Medium · the standard question

7 PROBLEMS

1. P two dice: total 8 or a double

ANSWER $\frac{5}{18}$

2. 5 red, 4 green, two drawn without replacement: $P(RR)$

ANSWER $\frac{5}{18}$

3. Same: P same colour

ANSWER $\frac{4}{9}$

4. Same: P different colours

ANSWER $\frac{5}{9}$

5. 50 pupils, 32 physics, 21 chemistry, 15 both: P neither

ANSWER 0.24

6. Same: $P(\text{physics} \mid \text{chemistry})$

ANSWER $\frac{5}{7}$

7. Same: $P(\text{chemistry} \mid \text{physics})$

ANSWER $\frac{15}{32}$

Hard · stretch and reason

7 PROBLEMS

1. P at least two heads in four throws

ANSWER $\frac{11}{16}$

2. Three cards from 52 without replacement: P all hearts

ANSWER $\frac{11}{850}$

3. Two dice: $P(\text{total } 8 \mid \text{first is even})$

ANSWER $\frac{1}{6}$

4. A bag of 6 white and 4 black: P the second is white

ANSWER $\frac{3}{5}$

5. A point in a circle of radius 5: P it lies within 3 of the centre

ANSWER 0.36

6. If $P(A) = 0.7$, $P(B) = 0.4$, $P(A \cup B) = 0.82$, are they independent?

ANSWER Yes — $P(A \cap B) = 0.28 = 0.7 \times 0.4$

7. A die is thrown until a six appears: P it takes at most three throws

ANSWER $\frac{91}{216} \approx 0.421$

07 Homework

Homework — 4 tasks

Bring the decision tree to Lesson 99; the revision block assumes it is done.

1. Rewrite in full every control-work question that lost a mark, naming the slip in the margin.
2. Copy the decision-tree table and add one worked example of your own to each row.

3. A bag holds 7 blue and 3 yellow counters; three are drawn without replacement. Find the probability that all three are blue.
4. Write your target for the revision block in one checkable sentence, and date it.

Sequences and progressions

[Cambridge revision]

Two formulas for the term, two for the sum — and one question that decides which pair.

LESSON 99–100

2 × 40 MIN

REVISION OF GRADE 9, §§18–21

P1 7.1–7.3

LESSON CLOCK

15'

25'

25'

25'

15'

5

Sequences and recurrences	15 min
Arithmetic progressions	25 min
Geometric progressions	25 min
Sums	25 min
Choosing the right pair	15 min
Homework	5 min

LEARNING OBJECTIVES

- Define a sequence by a formula for the n th term and by a recurrence.
- Identify an arithmetic or a geometric progression from its terms.
- Use the n th-term and sum formulas for both progressions.
- Solve problems where the number of terms is the unknown.

TEXTBOOK

National Algebra 9, pp. 190–214

Cambridge Pure Mathematics 1, pp. 148–162

Workbook Exercise 7A–7C

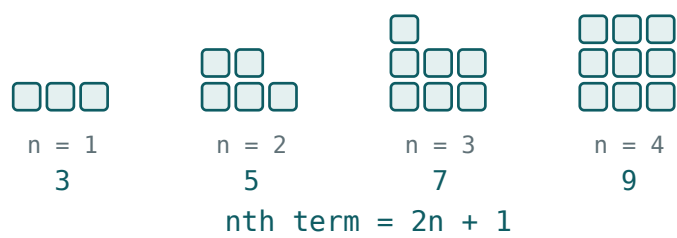
01

Explanation

Sequences and recurrences

A sequence can be given in two ways: by a **formula for the n th term**, which lets you jump straight to u_{100} , or by a **recurrence**, which tells you how to get from one term to the next.

SEQUENCE	NTH TERM	RECURRENCE
3, 7, 11, 15, ...	$u_n = 4n - 1$	$u_{n+1} = u_n + 4, u_1 = 3$
2, 6, 18, 54, ...	$u_n = 2 \cdot 3^{n-1}$	$u_{n+1} = 3u_n, u_1 = 2$
1, 4, 9, 16, ...	$u_n = n^2$	$u_{n+1} = u_n + 2n + 1$
1, 1, 2, 3, 5, ...	no simple formula	$u_{n+2} = u_{n+1} + u_n$



An arithmetic sequence plotted — the points lie on a straight line, and d is its gradient.

SIGMA NOTATION

$$\sum_{k=1}^n u_k = u_1 + u_2 + \dots + u_n$$

The letter under the sigma is a dummy: $\sum k^2$ and $\sum j^2$ are the same number.

Arithmetic progressions

Each term exceeds the one before by a fixed d .

$$u_n = a + (n - 1)d$$

The $n - 1$ matters: you take d one fewer times than the number of terms, because the first term is already there.

PROGRESSION	A	D	U_{10}
3, 7, 11, ...	3	4	39
20, 17, 14, ...	20	-3	-7
$\frac{1}{2}, 1, \frac{3}{2}, \dots$	$\frac{1}{2}$	$\frac{1}{2}$	5

TWO TERMS GIVE A AND D

From $u_4 = 11$ and $u_9 = 26$: subtracting gives $5d = 15$, so $d = 3$ and $a = 11 - 3(3) = 2$. Always subtract the two equations — never solve them one at a time.

Geometric progressions

Each term is a fixed multiple r of the one before.

$$u_n = a r^{(n-1)}$$

PROGRESSION	A	R	U_6
2, 6, 18, ...	2	3	486
16, 8, 4, ...	16	$\frac{1}{2}$	$\frac{1}{2}$
1, -2, 4, -8, ...	1	-2	-32

A NEGATIVE RATIO ALTERNATES IN SIGN

With $r = -2$ the terms swing above and below zero. Any question mentioning “alternating” or giving terms of opposite signs is telling you $r < 0$.

Two terms give a and r by dividing, not subtracting: from $u_3 = 12$ and $u_6 = 96$, $r^3 = 8$ so $r = 2$ and $a = 3$.

Sums

$$S_n = \frac{n}{2}(2a + (n-1)d) = \frac{n}{2}(a + l)$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

The second form of the arithmetic sum — *half the number of terms times the sum of the first and last* — is the fastest when the last term is known. The two geometric forms are identical; use whichever avoids negative signs.

QUESTION	USE	ANSWER
$3 + 7 + 11 + \dots$ (20 terms)	$\frac{n}{2}(2a + (n-1)d)$	820
$1 + 2 + 3 + \dots + 100$	$\frac{n}{2}(a + l)$	5050
$2 + 6 + 18 + \dots$ (8 terms)	$\frac{a(r^n - 1)}{r - 1}$	6560

Choosing the right pair

ONE TEST SETTLES IT

Take three consecutive terms. If the **differences** are equal it is arithmetic; if the **ratios** are equal it is geometric. If neither, it is neither — and the question wants something else.

When the unknown is n , the formula becomes an equation to solve — linear or quadratic for an AP, exponential for a GP.

QUESTION	EQUATION	ANSWER
how many terms of 3, 7, 11, ... sum to 820?	$\frac{n}{2}(6 + 4(n-1)) = 820$	$n = 20$
which term of 5, 8, 11, ... equals 98?	$5 + 3(n-1) = 98$	$n = 32$
which term of 3, 6, 12, ... first exceeds 1000?	$3 \cdot 2^{n-1} > 1000$	$n = 10$

N MUST BE A POSITIVE WHOLE NUMBER

A quadratic in n usually has one root that is negative or fractional. Reject it with a sentence, do not leave it in the answer.

02 Worked examples

EXAMPLE 1 An AP has $u_4 = 11$ and $u_9 = 26$. Find a , d and S_{20} .

$$(1) \quad a + 3d = 11, \quad a + 8d = 26$$

$$(2) \quad 5d = 15 \Rightarrow d = 3$$

Subtract.

$$(3) \quad a = 2$$

$$(4) \quad S_{20} = 10(4 + 57) = 610$$

ANSWER

$$a = 2, d = 3, S_{20} = 610$$

EXAMPLE 2 A GP has $u_3 = 12$ and $u_6 = 96$. Find a , r and S_8 .

$$(1) \quad \frac{ar^5}{ar^2} = \frac{96}{12} = 8$$

Divide.

$$(2) \quad r^3 = 8 \Rightarrow r = 2$$

$$(3) \quad ar^2 = 12 \Rightarrow a = 3$$

$$(4) \quad S_8 = \frac{3(2^8 - 1)}{1} = 765$$

ANSWER

$$a = 3, r = 2, S_8 = 765$$

EXAMPLE 3 How many terms of $3 + 7 + 11 + \dots$ are needed to reach a sum of 820?

$$① \quad \frac{n}{2}(6 + 4(n - 1)) = 820$$

$$② \quad n(4n + 2) = 1640$$

$$③ \quad 2n^2 + n - 820 = 0$$

$$④ \quad n = 20 \text{ (the root } -20.5 \text{ is rejected).}$$

ANSWER

$$n = 20$$

EXAMPLE 4 The first three terms of a GP are x , $x + 4$, $3x$. Find x .

$$① \quad \frac{x + 4}{x} = \frac{3x}{x + 4}$$

Equal ratios.

$$② \quad (x + 4)^2 = 3x^2$$

$$③ \quad 2x^2 - 8x - 16 = 0 \Rightarrow x^2 - 4x - 8 = 0$$

$$④ \quad x = 2 \pm 2\sqrt{3}$$

ANSWER

$$x = 2 + 2\sqrt{3} \text{ or } x = 2 - 2\sqrt{3}$$

03 Interactive model

Write two mystery sequences on the board and have the class decide AP, GP or neither by differences and ratios alone.

Arithmetic, geometric, or neither?

3, 7, 11, 15 is:

AP

GP

neither

both

2, 6, 18, 54 is:

AP

GP

neither

both

1, 4, 9, 16 is:

AP

GP

neither

both

u_{10} of 3, 7, 11, ...:

39

43

40

36

u_6 of 16, 8, 4, ...:

1

$\frac{1}{2}$

2

$\frac{1}{4}$

$1 + 2 + \dots + 100$ is:

5000

5050

10100

4950

To find r from two terms you:

subtract

divide

add

square

A negative r makes the terms:

grow

shrink

alternate in sign

constant

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Sequence	Ketma-ketlik	Последовательность
Term	Had	Член
Recurrence relation	Rekurrent munosabat	Рекуррентное соотношение
Arithmetic progression	Arifmetik progressiya	Арифметическая прогрессия
Geometric progression	Geometrik progressiya	Геометрическая прогрессия
Common difference	Ayirma	Разность
Common ratio	Maxraj	Знаменатель
Sum of n terms	n ta hadning yig'indisi	Сумма n членов
Sigma notation	Sigma belgisi	Знак суммы
Convergent	Yaqinlashuvchi	Сходящаяся

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself u_n of an AP is:

$a + nd$

$a + (n - 1)d$

ar^n

$a + (n + 1)d$

 S_n of an AP is:

$\frac{n}{2}(a + l)$

$n(a + l)$

na

$\frac{a}{1 - r}$

 u_n of a GP is:

ar^n

$ar^{(n-1)}$

$a + r^n$

anr

The GP sum formula fails when:

$$r = 0$$

$$r = 1$$

$$r < 0$$

$$a = 0$$

Three terms are geometric when:

the differences are equal

the ratios are equal

they are positive

the middle is the mean

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. u_{10} of 3, 7, 11, ...

ANSWER 39

2. u_{15} of 20, 17, 14, ...

ANSWER -22

3. u_5 of 2, 6, 18, ...

ANSWER 162

4. u_6 of 16, 8, 4, ...

ANSWER $\frac{1}{2}$

5. S_{10} of 1, 2, 3, ...

ANSWER 55

6. S_5 of 3, 6, 12, ...

ANSWER 93

7. Is 5, 10, 20, 40 an AP or a GP?

ANSWER GP, $r = 2$

Medium · the standard question

7 PROBLEMS

1. AP with $u_4 = 11$, $u_9 = 26$: find a and d

ANSWER $a = 2, d = 3$

2. Same AP: S_{20}

ANSWER 610

3. GP with $u_3 = 12$, $u_6 = 96$: find a and r

ANSWER $a = 3, r = 2$

4. Same GP: S_8

ANSWER 765

5. Which term of $5, 8, 11, \dots$ equals 98 ?

ANSWER $n = 32$

6. $1 + 2 + 3 + \dots + 100$

ANSWER 5050

7. Sum of the first 20 terms of $3, 7, 11, \dots$

ANSWER 820

Hard · stretch and reason

7 PROBLEMS

1. How many terms of $3 + 7 + 11 + \dots$ give 820?

ANSWER $n = 20$

2. First three terms of a GP are $x, x + 4, 3x$: find x

ANSWER $2 \pm 2\sqrt{3}$

3. Which term of $3, 6, 12, \dots$ first exceeds 1000?

ANSWER $n = 10$

4. An AP has $S_{10} = 155$ and $S_{20} = 610$: find a and d

ANSWER $a = 2, d = 3$

5. The sum of the first n odd numbers

ANSWER n^2

6. A GP has $u_2 = 6$ and $u_5 = -48$: find r

ANSWER $r = -2$

7. Three numbers in AP have sum 15 and product 80: find them

ANSWER 2, 5, 8 (or reversed)

07 Homework

Homework — 5 tasks

Write a and d (or r) on their own line before every calculation.

1. An AP has $u_5 = 17$ and $u_{12} = 45$. Find a, d and S_{25} .
2. A GP has $u_2 = 10$ and $u_5 = 80$. Find a, r and S_7 .
3. How many terms of $2 + 5 + 8 + \dots$ are needed for a sum over 1000?
4. The numbers $x - 2, x, x + 6$ form a GP. Find x .

5. Show that the sum of the first n even numbers is $n(n + 1)$, and check it for $n = 6$.

Infinite geometric progressions and the binomial expansion

[Cambridge revision]

A sum that never ends but still has a value, and a triangle that expands any bracket.

LESSON 101–102

2 × 40 MIN

REVISION OF GRADE 9, §22

P1 7.4, 6.1–6.3

LESSON CLOCK



When an infinite sum exists 20 min

Recurring decimals 20 min

Pascal's triangle 25 min

The binomial formula 25 min

One term without the rest 15 min

Homework 5 min

LEARNING OBJECTIVES

- State when an infinite geometric series converges, and find its sum.
- Convert a recurring decimal to a fraction with the sum to infinity.
- Expand $(a + b)^n$ using Pascal's triangle and the binomial coefficients.
- Find one named term of an expansion without writing the others.

TEXTBOOK

National Algebra 9, pp. 215–228

Cambridge Pure Mathematics 1, pp. 128–147, 163–168

Workbook Exercise 6A–6C, 7D

01 Explanation

When an infinite sum exists

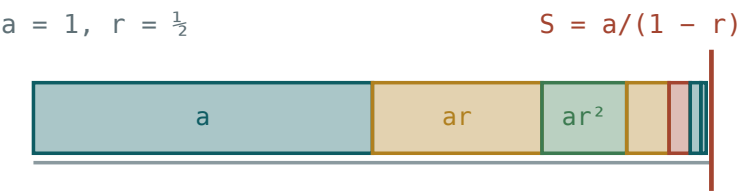
Add $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$ and the total never passes 2. Add $1 + 2 + 4 + 8 + \dots$

and it passes everything. The difference is the ratio.

$$S_{\infty} = \frac{a}{1 - r}, \text{ valid exactly when } |r| < 1$$

Why: in $S_n = \frac{a(1 - r^n)}{1 - r}$, the term r^n shrinks to zero when $|r| < 1$ and grows without

limit otherwise.



the pieces never reach the wall, and never pass it

Each piece is half the last — the pieces approach the wall and never reach it.

|R| < 1 IS THE WHOLE CONDITION

$r = 1$ gives $a + a + a + \dots$, which diverges; $r = -1$ gives $a - a + a - \dots$, which oscillates and has no sum. Both are excluded, and a question that asks “for what x does the series converge” is asking you to solve $|r| < 1$.

Recurring decimals

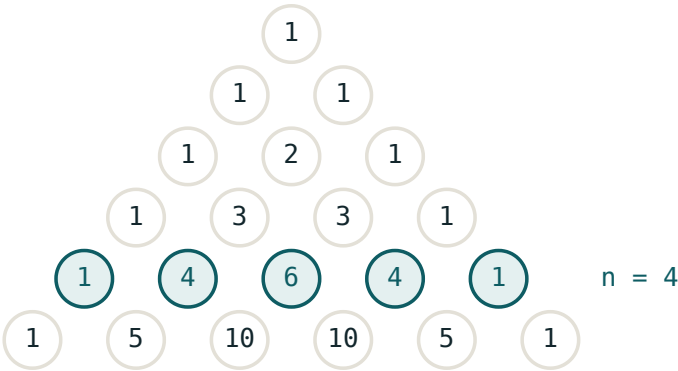
Every recurring decimal is a geometric series in disguise, so every one is a fraction.

DECIMAL	AS A SERIES	A	R	FRACTION
0.333...	$0.3 + 0.03 + \dots$	0.3	0.1	$\frac{1}{3}$
0.272727...	$0.27 + 0.0027 + \dots$	0.27	0.01	$\frac{3}{11}$
0.999...	$0.9 + 0.09 + \dots$	0.9	0.1	1

The last line is not a trick: $\frac{0.9}{1 - 0.1} = \frac{0.9}{0.9} = 1$. The decimal 0.999... **is** 1, written differently.

Pascal’s triangle

Expand $(a + b)^n$ for small n and the coefficients form a triangle in which every entry is the sum of the two above it.



each entry is the sum of the two above it

Row n gives the coefficients of $(a + b)^n$; the row highlighted is $n = 4$.

N	COEFFICIENTS	EXPANSION
2	1 2 1	$a^2 + 2ab + b^2$
3	1 3 3 1	$a^3 + 3a^2b + 3ab^2 + b^3$
4	1 4 6 4 1	$a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$
5	1 5 10 10 5 1	$a^5 + 5a^4b + \dots + b^5$

TWO PATTERNS TO CHECK EVERY EXPANSION AGAINST

The powers of a fall from n to 0 while those of b rise from 0 to n , so **every term's powers add to n** . And there are always $n + 1$ terms.

The binomial formula

$$(a + b)^n = \sum_{k=0}^n C(n, k) a^{n-k} b^k$$

$$C(n, k) = \frac{n!}{k!(n - k)!}$$

The coefficient $C(n, k)$ — written nC_k or $\binom{n}{k}$ — is exactly the entry in row n , position k , of Pascal's triangle. For n up to about 6 the triangle is faster; beyond that, the factorial formula is.

$C(N, K)$	VALUE	WHY
$C(5, 0)$	1	one way to choose nothing
$C(5, 1)$	5	$= n$
$C(5, 2)$	10	$\frac{5 \times 4}{2}$
$C(5, 3)$	10	$= C(5, 2)$
$C(8, 3)$	56	$\frac{8 \times 7 \times 6}{3 \times 2 \times 1}$

THE SYMMETRY THAT HALVES THE WORK

$$C(n, k) = C(n, n - k)$$

Choosing k to include is the same as choosing $n - k$ to leave out. So $C(20, 18) = C(20, 2) = 190$ — three multiplications instead of eighteen.

One term without the rest

Examinations rarely want the whole expansion. They want one coefficient.

the term in b^k is $C(n, k) a^{n-k} b^k$

Example. The coefficient of x^3 in $(2 + x)^5$: here $k = 3$, $a = 2$, so the term is $C(5, 3) \cdot 2^2 \cdot x^3 = 10 \times 4 x^3 = 40x^3$.

With a coefficient on x , raise it too. In $(1 + 3x)^4$, the term in x^2 is $C(4, 2)(3x)^2 = 6 \times 9x^2 = 54x^2$ — the 3 is squared as well.

A MINUS SIGN BELONGS TO THE WHOLE BRACKET

In $(1 - 2x)^5$ write $b = -2x$, so the term in x^3 is $C(5, 3)(-2x)^3 = 10(-8x^3) = -80x^3$. Forgetting to cube the minus is the single commonest error in this topic.

02 Worked examples

EXAMPLE 1 Find the sum to infinity of $12 + 4 + \frac{4}{3} + \dots$

$$\textcircled{1} \quad r = \frac{4}{12} = \frac{1}{3}$$

$|r| < 1$ — it converges.

$$\textcircled{2} \quad S_{\infty} = \frac{12}{1 - \frac{1}{3}}$$

$$\textcircled{3} \quad = \frac{12}{\frac{2}{3}} = 18$$

ANSWER

18

EXAMPLE 2 Write $0.272727\dots$ as a fraction.

$$\textcircled{1} \quad a = 0.27, r = 0.01$$

$$\textcircled{2} \quad S = \frac{0.27}{0.99}$$

$$\textcircled{3} \quad = \frac{27}{99} = \frac{3}{11}$$

ANSWER

$\frac{3}{11}$

EXAMPLE 3 Expand $(2 + x)^4$ fully.

① Row 4: 1, 4, 6, 4, 1.

② $2^4 + 4(2^3)x + 6(2^2)x^2 + 4(2)x^3 + x^4$

③ $16 + 32x + 24x^2 + 8x^3 + x^4$
Five terms, powers adding to 4.

ANSWER

$$16 + 32x + 24x^2 + 8x^3 + x^4$$

EXAMPLE 4 Find the coefficient of x^3 in $(1 - 2x)^6$.

① $C(6, 3) = 20$

② $(-2x)^3 = -8x^3$
Cube the whole bracket.

③ $20 \times (-8) = -160$

ANSWER

$$-160$$

03 Interactive model

Cut a strip of paper in half repeatedly and lay the pieces end to end — the sum to infinity appears as a finite length.

Convergence, and coefficients

S_{∞} exists when:

S_{∞} of $1 + \frac{1}{2} + \frac{1}{4} + \dots$:

0.999... equals:

Row 4 of Pascal's triangle:

1 3 3 1

1 4 6 4 1

1 5 10 10 5 1

1 4 4 1

$(a + b)^5$ has how many terms?

5

6

7

10

$C(8, 3)$ is:

24

56

336

112

$C(20, 18)$ equals:

$C(20, 2)$

$C(20, 18)$ only

20

180

The term in x^3 of $(1 - 2x)^6$:

$$160x^3$$

$$-160x^3$$

$$-20x^3$$

$$-8x^3$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Infinite series	Cheksiz qator	Бесконечный ряд
Sum to infinity	Cheksiz yig'indi	Сумма бесконечной прогрессии
Convergent	Yaqinlashuvchi	Сходящийся
Divergent	Uzoqlashuvchi	Расходящийся
Recurring decimal	Davriy o'nli kasr	Периодическая дробь
Binomial expansion	Binomial yoyilma	Разложение бинома
Binomial coefficient	Binomial koeffitsiyent	Биномиальный коэффициент
Pascal's triangle	Paskal uchburchagi	Треугольник Паскаля
Factorial	Faktorial	Факториал
General term	Umumiy had	Общий член

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

An infinite GP converges when:

$$|r| < 1$$

$$r > 1$$

$$a > 0$$

always

S_{∞} equals:

$$\frac{a}{1+r}$$

$$\frac{a}{1-r}$$

$$\frac{a}{r}$$

$$ar$$

Every entry of Pascal's triangle is:

a prime

the sum of the two above

a square

random

In every term of $(a + b)^n$ the powers:

add to n

are equal

add to $2n$

add to k

$C(n, k) = C(n, n - k)$ because:

of the formula only

choosing k in is choosing $n - k$ out

it is a coincidence

of Pascal

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. S_{∞} of $1 + \frac{1}{2} + \frac{1}{4} + \dots$

ANSWER 2

2. S_{∞} of $9 + 3 + 1 + \dots$

ANSWER 13.5

3. Does $1 + 2 + 4 + \dots$ converge?

ANSWER No — $r = 2$

4. $0.333\dots$ as a fraction

ANSWER $\frac{1}{3}$

5. Row 3 of Pascal's triangle

ANSWER 1 3 3 1

6. Number of terms in $(a + b)^7$

ANSWER 8

7. $C(6, 2)$

ANSWER 15

Medium · the standard question

7 PROBLEMS

1. S_{∞} of $12 + 4 + \frac{4}{3} + \dots$

ANSWER 18

2. S_{∞} of $8 - 4 + 2 - \dots$

ANSWER $\frac{16}{3}$

3. $0.272727\dots$ as a fraction

ANSWER $\frac{3}{11}$

4. Expand $(1 + x)^4$

ANSWER $1 + 4x + 6x^2 + 4x^3 + x^4$

5. Expand $(2 + x)^4$

ANSWER $16 + 32x + 24x^2 + 8x^3 + x^4$

6. Coefficient of x^2 in $(1 + 3x)^4$

ANSWER 54

7. $C(8, 3)$

ANSWER 56

Hard · stretch and reason

7 PROBLEMS

1. Coefficient of x^3 in $(1 - 2x)^6$

ANSWER -160

2. Coefficient of x^4 in $(2 - x)^7$

ANSWER 280

3. Constant term in $(x + \frac{2}{x})^6$

ANSWER 160

4. For which x does $1 + 2x + 4x^2 + \dots$ converge, and to what?

ANSWER $|x| < \frac{1}{2}; \frac{1}{1 - 2x}$

5. A GP has $S_\infty = 27$ and $a = 18$: find r

ANSWER $r = \frac{1}{3}$

6. A ball dropped from 10 m rebounds 0.6 of its height each time: total distance travelled

ANSWER 40 m

7. Show that $C(n, k) + C(n, k + 1) = C(n + 1, k + 1)$

ANSWER Pascal's rule, from the factorial form

07 Homework

Homework — 5 tasks

This is the last algebra lesson of Grade 10; keep the summary page for Grade 11.

- Find the sum to infinity of $20 + 5 + 1.25 + \dots$, and state the value of r .
- Write $0.4545\dots$ and $0.1666\dots$ as fractions using the sum to infinity.
- Expand $(3 - x)^4$ fully, and check that the powers in each term add to 4.
- Find the coefficient of x^3 in $(2 + 3x)^5$ without writing the other terms.

5. A GP has sum to infinity 12 and first term 8. Find r and the fourth term.